

Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)

CS423 / CSC13003 – Software Testing (AI-augmented · 2026)

AI Audit Report – HW06-AI API Testing

1. Student Information

Field	Value
Student name (printed):	Nguyen Bao Duy
Student ID:	23127179
Class / Cohort:	23KTPM2
Assignment ID:	HW06-AI
Assignment date:	22/08/2026
AI tool(s) used:	Codex (FR-04 and FR-15 workflow phases), Claude Opus 5 (<code>claude-opus-5</code>) via Claude Code (FR-08 and FR-15 generation batches B1-B5), and OpenAI Codex (GPT-5) for FR-15 generation batch B6
AI tool(s) used:	☒ Yes ☐ No

2. Instructions and audit evidence status

This report follows the FIT@HCMUS AI Audit Report template. It contains one row per original AI-generated test-case artifact and one row for each later major AI-generated artifact bundle that has recoverable prompt context, output evidence, and student review. It is cumulative through the final-submission curation interaction and covers FR-04 `PUT /api/users/me`, FR-08 `POST /api/checkout`, FR-15 `POST /api/products`, and the final report/critique/generator/evidence bundle. The canonical source preserves 47 FR-04, 57 FR-08, and 59 FR-15 AI test artifacts, and the separate human-review registry preserves their confirmed verdicts, reasoning, and corrections. The three API phases differ in provenance quality and the difference is disclosed rather than smoothed over: FR-04 prompts were not captured at the time, whereas every FR-08 and FR-15 candidate stores its verbatim bounded prompt, batch id, actual tool, and real generation timestamp.

The report does not add test-artifact rows for the seven FR-04, five FR-08, or five FR-15 HUMAN extension cases, because they are student-selected or student-authored extensions rather than AI-generated test candidates. Later targeted Newman runs are recorded as evidence of AI-assisted correction and triage; they do not themselves constitute human bug confirmation or GitHub Issue publication.

For FR-04, the exact standalone generation prompt was not separately captured in the repository. Each FR-04 test-case row therefore uses the preserved artifact-level generation context and actual tool (Codex) and explicitly marks the limitation; no prompt has been reconstructed from the output. FR-08 and FR-15 test candidates instead use their stored verbatim bounded prompts and timestamps. Later workflow rows use actual user-turn excerpts when recoverable and explicitly mark unavailable standalone prompts, exact models, or chat times rather than inferring them from Git commits.

Historical FR-04 interaction times that were never captured remain explicitly unavailable rather than guessed. The official HW06 specification does not require a handwritten/electronic signature, and it lists a teaching team rather than one instructor; those template fields are completed accordingly below.

3. Audit Table – one row per artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Artifact context: Guided FR-04 generation: dimensions and domain partitions. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-001; title=Authenticated ordinary profile update; preconditions=Known user JWT and captured baseline profile; request=PUT /api/users/me with ordinary name, address, and 10-digit phone 0123456789; expected=Update is accepted under the documented editable-field contract; exact status/body is SPEC GAP.	VALID	Accepted as an ordinary authenticated update of the documented editable fields. Evidence basis: official_contract; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: dimensions and domain partitions. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-002; title=Unicode profile text; preconditions=Known user JWT and captured baseline profile; request=PUT with Vietnamese Unicode name/address and valid phone; expected=Editable text can be submitted; exact response is SPEC GAP.	INVALID	Duplicate/overlap with FR04-AI-019 and FR04-AI-043; Unicode support was also stated too strongly. Evidence basis: official_contract_plus_exploration; requirement references: FR-04, SEC-04.	Rejected from the reviewed executable suite; covered by FR04-AI-019.
Artifact context: Guided FR-04 generation: dimensions and domain partitions. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-003; title=Phone length boundary map; preconditions=Known user JWT; request=Run phone values with 9, 10, 11, and 12 digits, preserving a leading zero; expected=10- and 11-digit leading-zero values are contract-valid; invalid outcomes/statuses are SPEC GAP.	INVALID	Compound boundary map duplicates individual FR04-AI-011 through FR04-AI-014. Evidence basis: official_contract; requirement references: FR-04.	Rejected from the reviewed executable suite; covered by FR04-AI-011 through FR04-AI-014.
Artifact context: Guided FR-04 generation: dimensions and domain partitions. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-004; title=Single-field update omission behavior; preconditions=Known user JWT and baseline profile; request=Update only one documented profile field; expected=Observe omitted-field behavior; requiredness and preservation are SPEC GAP.	INCOMPLETE	The original candidate does not identify which single editable field is updated, so the request and observable state comparison are not deterministic. Evidence basis: exploratory; requirement references: FR-04.	Make deterministic: update only phone to 0123456789, then GET /api/users/me and observe whether omitted name and shipping_address remain unchanged; omission semantics remain SPEC GAP.
Artifact context: Guided FR-04 generation: authentication and JWT. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-005; title=Missing Authorization header; preconditions=No Authorization header; request=Send otherwise valid profile update without Authorization; expected=Request requires a valid JWT under SEC-02; failure status/schema is SPEC GAP.	VALID	Meaningful missing-JWT authentication partition under SEC-02. Evidence basis: official_contract; requirement references: FR-04, SEC-02.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: authentication and JWT. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-006; title=Basic authentication scheme; preconditions=Authorization uses Basic scheme; request=Send Authorization: Basic ... with otherwise valid body; expected=Request	VALID	Meaningful wrong-authentication-scheme partition under SEC-02. Evidence basis: official_contract; requirement references: FR-04, SEC-02.	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
	requires Bearer JWT; exact failure status/schema is SPEC GAP.			
Artifact context: Guided FR-04 generation: authentication and JWT. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-007; title=Empty Bearer token; preconditions=Authorization header is Bearer with no token; request=Send Authorization: Bearer and valid body; expected=Request requires a valid JWT; exact failure status/schema is SPEC GAP.	VALID	Meaningful empty-Bearer-token authentication partition under SEC-02. Evidence basis: official_contract; requirement references: FR-04, SEC-02.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: authentication and JWT. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-008; title=Invalid-signature JWT; preconditions=JWT-like token with invalid signature; request=Send invalid-signature Bearer token and valid body; expected=Request requires a valid JWT; exact failure status/schema is SPEC GAP.	VALID	Meaningful invalid-signature JWT partition under SEC-02. Evidence basis: official_contract; requirement references: FR-04, SEC-02.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: authentication and JWT. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-009; title=Expired JWT; preconditions=Expired user JWT; request=Send expired Bearer token and valid body; expected=Request requires a valid JWT; exact failure status/schema is SPEC GAP.	VALID	Meaningful expired-JWT authentication partition under SEC-02. Evidence basis: official_contract; requirement references: FR-04, SEC-02.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: authentication and JWT. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-010; title=Valid JWT isolates user update; preconditions=Valid user JWT and known user identity; request=Update profile, then GET /api/users/me; expected=Update affects the authenticated user; read-back verifies the state.	INCOMPLETE	The original candidate claims user isolation without performing a second-user comparison, so the assertion is broader than the evidence collected by the case. Evidence basis: official_contract; requirement references: FR-04, SEC-02.	With a valid user JWT, update the profile and GET /api/users/me; verify the returned authenticated user's editable fields match the submitted update. Do not claim cross-user isolation without a second-user comparison.
Artifact context: Guided FR-04 generation: phone partitions and boundaries. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-011; title=Ten-digit leading-zero phone; preconditions=Valid user JWT; request=phone=0123456789; expected=Accepted as a documented valid phone partition; exact status/body is SPEC GAP.	VALID	Covers the documented 10-digit leading-zero valid phone boundary. Evidence basis: official_contract; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: phone partitions and boundaries. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-012; title=Eleven-digit leading-zero phone; preconditions=Valid user JWT; request=phone=01234567890; expected=Accepted as a documented valid phone partition; exact status/body is SPEC GAP.	VALID	Covers the documented 11-digit leading-zero valid phone boundary. Evidence basis: official_contract; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: phone partitions and boundaries. Tool: Codex. Exact	Verbatim registry artifact excerpt: ID=FR04-AI-013; title=Nine-digit phone; preconditions=Valid user	INCOMPLETE	The 9-digit value violates the documented 10/11-digit phone rule, but the original oracle does not clearly state	Use phone=012345678. It violates the documented 10/11-digit rule and must not become the user's valid

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standalone prompt text was not separately captured; this is the preserved artifact-level source context.	JWT; request=phone=012345678; expected=Outside documented 10-11 digit rule; rejection behavior/status is SPEC GAP.		the persistence invariant for the invalid value. Evidence basis: official_contract; requirement references: FR-04.	persisted phone value. Exact HTTP status and error schema remain SPEC GAP.
Artifact context: Guided FR-04 generation: phone partitions and boundaries. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-014; title=Twelve-digit phone; preconditions=Valid user JWT; request=phone=0123456789 01; expected=Outside documented 10-11 digit rule; rejection behavior/status is SPEC GAP.	INCOMPLETE	The 12-digit value violates the documented 10/11-digit phone rule, but the original oracle is too weak about persistence of an invalid value. Evidence basis: official_contract; requirement references: FR-04.	Use phone=012345678901. It violates the documented 10/11-digit rule and must not become the user's valid persisted phone value. Exact HTTP status and error schema remain SPEC GAP.
Artifact context: Guided FR-04 generation: phone partitions and boundaries. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-015; title=Phone without leading zero; preconditions=Valid user JWT; request=phone=1123456789 ; expected=Violates documented leading-zero rule; rejection behavior/status is SPEC GAP.	INCOMPLETE	The phone value violates the documented leading-zero rule, but the original oracle does not clearly express the persisted-state invariant. Evidence basis: official_contract; requirement references: FR-04.	Use phone=1123456789. It violates the documented leading-zero rule and must not become the user's valid persisted phone value. Exact HTTP status and error schema remain SPEC GAP.
Artifact context: Guided FR-04 generation: phone partitions and boundaries. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-016; title=Phone containing a letter; preconditions=Valid user JWT; request=phone=012345678A ; expected=Violates documented digit format; rejection behavior/status is SPEC GAP.	INCOMPLETE	The phone contains a non-digit character and violates the documented format, but the original oracle is not sufficiently concrete for later execution. Evidence basis: official_contract; requirement references: FR-04.	Use phone=012345678A. It violates the documented digit-format rule and must not become the user's valid persisted phone value. Exact HTTP status and error schema remain SPEC GAP.
Artifact context: Guided FR-04 generation: phone partitions and boundaries. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-017; title=International phone format; preconditions=Valid user JWT; request=phone=+841234567 89; expected=Does not use the documented leading-zero format; rejection behavior/status is SPEC GAP.	INCOMPLETE	The +84 representation does not satisfy the documented leading-zero format, but the original oracle is too vague about the invalid persisted state. Evidence basis: official_contract; requirement references: FR-04.	Use phone=+84123456789. It does not satisfy the documented phone format and must not become the user's valid persisted phone value. Exact HTTP status and error schema remain SPEC GAP.
Artifact context: Guided FR-04 generation: name and shipping address partitions. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-018; title=Ordinary ASCII name and address; preconditions=Valid user JWT; request=Submit ordinary ASCII name/address and valid phone; expected=Editable text update is supported; exact response is SPEC GAP.	VALID	Meaningful ordinary ASCII text/domain partition. Evidence basis: official_contract; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: name and shipping address partitions. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-019; title=Vietnamese Unicode name and address; preconditions=Valid user JWT; request=Submit Vietnamese Unicode name/address and valid phone; expected=Observe	VALID	Meaningful Vietnamese Unicode text/domain partition. Evidence basis: official_contract_plus_exploration; requirement references: FR-04, SEC-04.	Accepted as-is; no correction required.

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	text persistence; exact response is SPEC GAP.			
Artifact context: Guided FR-04 generation: name and shipping address partitions. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-020; title=Empty name robustness; preconditions=Valid user JWT; request=Submit empty name with otherwise valid fields; expected=Requiredness is undocumented; observe behavior without requiring a status.	VALID	Useful exploratory empty-name robustness partition without inventing a mandatory rejection. Evidence basis: exploratory; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: name and shipping address partitions. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-021; title=Null shipping address robustness; preconditions=Valid user JWT; request=Submit null shipping_address with otherwise valid fields; expected=Nullability and status are SPEC GAP; observe behavior.	VALID	Useful exploratory null shipping-address robustness partition without inventing a mandatory rejection. Evidence basis: exploratory; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: name and shipping address partitions. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-022; title=Wrong types for text fields; preconditions=Valid user JWT; request=Submit numeric name and array shipping_address; expected=Wrong-type robustness behavior is unspecified; status is SPEC GAP.	INCOMPLETE	The candidate combines two wrong-type fields in one request, creating fault masking and preventing isolated diagnosis. Evidence basis: exploratory; requirement references: FR-04.	Submit numeric name with otherwise valid shipping_address and phone. Name type handling is unspecified; observe behavior without inventing a mandatory HTTP status. Do not also submit an array shipping_address.
Artifact context: Guided FR-04 generation: name and shipping address partitions. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-023; title=Very long profile text; preconditions=Valid user JWT; request=Submit very long name and address; expected=No length limit is documented; observe behavior without requiring rejection.	INCOMPLETE	The phrase "very long" does not define a deterministic test input. Evidence basis: exploratory; requirement references: FR-04.	Submit a name consisting of 256 characters with otherwise valid fields. No name length limit is documented; observe behavior without requiring rejection. Keep exploratory.
Artifact context: Guided FR-04 generation: name and shipping address partitions. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-024; title=HTML-like profile text; preconditions=Valid user JWT and UI/read-back access; request=Submit Lan in name/address and inspect the consuming display boundary; expected=SEC-04 requires safe escaping at display; API rejection is not required.	VALID	Valid SEC-04 display-boundary exploration without assuming API rejection. Evidence basis: exploratory; requirement references: FR-04, SEC-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: protected email and role fields. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-025; title=Attempt to change email; preconditions=Valid user JWT and baseline email; request=Submit a different email with valid editable fields; expected=Persisted email remains baseline; exact	INCOMPLETE	The original candidate states that email must remain unchanged but does not explicitly include persisted-state verification. Evidence basis: official_contract; requirement references: FR-04.	Capture baseline email; PUT a different email with otherwise valid editable fields; GET; verify persisted email still equals baseline. Exact handling/status of the protected property remains SPEC GAP.

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	extra-field handling is SPEC GAP.			
Artifact context: Guided FR-04 generation: protected email and role fields. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-026; title=Attempt to escalate role; preconditions=Valid user JWT and baseline role; request=Submit role=admin with valid editable fields; expected=Persisted role remains unchanged under SEC-06; exact handling/status is SPEC GAP.	INCOMPLETE	The original role-tampering candidate requires an explicit baseline and post-update persisted-role comparison. Evidence basis: official_contract; requirement references: FR-04, SEC-06.	Capture baseline role; PUT role="admin" with otherwise valid editable fields; GET; verify persisted role still equals baseline. Do not classify implementation behavior as a confirmed bug before execution.
Artifact context: Guided FR-04 generation: protected email and role fields. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-027; title=Change email and role together; preconditions=Valid user JWT and baseline protected fields; request=Submit changed email and role with valid editable fields; expected=Neither protected field changes; exact status/body is SPEC GAP.	INCOMPLETE	The combined email/role tampering idea is meaningful, but the original case does not explicitly define verification of both persisted protected fields. Evidence basis: official_contract; requirement references: FR-04, SEC-06.	Capture baseline email and role; PUT both changed protected fields plus valid editable fields; GET; verify both protected fields remain baseline. Exact PUT status/body remains SPEC GAP.
Artifact context: Guided FR-04 generation: protected email and role fields. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-028; title=Wrong-type protected fields; preconditions=Valid user JWT and baseline protected fields; request=Submit numeric email and object role; expected=No protected-field mutation; handling/status is exploratory and SPEC GAP.	INVALID	Combines wrong-type email and role even though both are protected fields; compound input adds little diagnostic value. Evidence basis: exploratory; requirement references: FR-04, SEC-06.	Rejected from the reviewed executable suite; covered by the separate protected-field cases.
Artifact context: Guided FR-04 generation: persistence and repeated state. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-029; title=Persist all editable fields; preconditions=Valid user JWT and baseline profile; request=Update all editable fields, then GET /api/users/me; expected=New name/address/phone persist for the authenticated user.	VALID	Verifies PUT followed by GET persistence of all editable fields. Evidence basis: official_contract; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: persistence and repeated state. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-030; title=Phone-only update preservation; preconditions=Valid user JWT and baseline profile; request=Update only phone, then GET profile; expected=Observe whether omitted fields remain baseline; omission semantics are SPEC GAP.	INCOMPLETE	The candidate does not specify a concrete phone value or the exact baseline fields to compare after the partial update. Evidence basis: exploratory; requirement references: FR-04.	Capture baseline name and shipping_address; PUT only phone=0123456789; GET; observe whether both remain baseline. Omission semantics are SPEC GAP, not an official preservation assertion.
Artifact context: Guided FR-04 generation: persistence and repeated state. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-031; title=Repeated V2 to V3 updates; preconditions=Valid user JWT and baseline profile; request=Update to distinct V2, update to distinct V3, then GET profile; expected=Final persisted profile is V3, not stale V2.	VALID	Verifies the repeated V2-to-V3 state transition and final persistence. Evidence basis: official_contract; requirement references: FR-04.	Accepted as-is; no correction required.

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Artifact context: Guided FR-04 generation: persistence and repeated state. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-032; title=Restore baseline after mutation; preconditions=Valid user JWT and captured baseline; request=Restore baseline fields after a mutation, then GET profile; expected=Original profile state is restored for test isolation.	INVALID	This is cleanup/test-isolation procedure, not an independent product test case. Evidence basis: state_management; requirement references: FR-04.	Move the baseline-restoration behavior to supporting cleanup/teardown procedure; do not count it as a test case.
Artifact context: Guided FR-04 generation: persistence and repeated state. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-033; title=Unauthenticated update does not mutate profile; preconditions=No valid JWT, plus valid session for subsequent GET; request=Attempt update without authentication, then GET with valid session; expected=SEC-02 blocks unauthenticated mutation; failure status is SPEC GAP and baseline remains unchanged.	VALID	Meaningfully tests unauthenticated mutation and authenticated read-back without execution claims. Evidence basis: official_contract; requirement references: FR-04, SEC-02.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: SEC-04, SEC-05, and SEC-06 exploration. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-034; title=Escaped HTML at display boundary; preconditions=Valid user JWT and access to consuming profile UI; request=Store HTML-like profile text and inspect rendered profile output; expected=User-controlled text is escaped at display boundary under SEC-04; API rejection is not required.	INVALID	Substantially duplicates FR04-AI-024 SEC-04 display-boundary test. Evidence basis: exploratory; requirement references: FR-04, SEC-04.	Rejected from the reviewed executable suite; covered by FR04-AI-024.
Artifact context: Guided FR-04 generation: SEC-04, SEC-05, and SEC-06 exploration. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-035; title=Quote and apostrophe persistence; preconditions=Valid user JWT; request=Store harmless quote/apostrophe text and GET profile; expected=Stored text remains intact; black-box result does not prove query implementation.	INCOMPLETE	The candidate does not provide a concrete quote/apostrophe payload or a sufficiently observable SEC-05 oracle. Evidence basis: exploratory; requirement references: FR-04, SEC-05.	Use concrete harmless payload name=O'Connor "Test". Observe unintended mutation, server failure, or corrupted persistence; a safe black-box result does not prove parameterized queries.
Artifact context: Guided FR-04 generation: SEC-04, SEC-05, and SEC-06 exploration. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-036; title=SQL-like text robustness; preconditions=Valid user JWT; request=Submit SQL-like text in name/address and GET profile; expected=Use as SEC-05 robustness exploration; safe response alone does not prove parameterized queries.	INCOMPLETE	The phrase "SQL-like text" is not executable without a concrete inert payload, and the observable black-box oracle must avoid claiming internal query parameterization. Evidence basis: exploratory; requirement references: FR-04, SEC-05.	Use inert SQL-like text Robert'); SELECT 1; --. Observe unintended mutation, server failure, or corrupted persistence; do not claim a normal response proves parameterized queries.

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Artifact context: Guided FR-04 generation: SEC-04, SEC-05, and SEC-06 exploration. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-037; title=Role invariant security check; preconditions=Valid user JWT and baseline role; request=Submit role-tampering payload and compare persisted role; expected=Persisted role equals baseline under SEC-06; no bug is assumed.	INVALID	Substantially duplicates FR04-AI-026 role-tampering invariant test. Evidence basis: official_contract; requirement references: FR-04, SEC-06.	Rejected from the reviewed executable suite; covered by FR04-AI-026.
Artifact context: Guided FR-04 generation: schema and request shape. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-038; title=Valid JSON object schema; preconditions=Valid user JWT; request=Send JSON object containing the three documented editable strings; expected=Editable fields can be updated and read back; response schema beyond this is SPEC GAP.	INCOMPLETE	The candidate is labeled as schema validation even though the exact PUT response schema is undocumented; it needs to be reframed around documented request fields and persisted read-back. Evidence basis: official_contract; requirement references: FR-04.	Send a JSON object containing documented name, shipping_address, and phone, then GET and verify those fields. Do not assert an undocumented exact PUT response schema.
Artifact context: Guided FR-04 generation: schema and request shape. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-039; title=Empty JSON object; preconditions=Valid user JWT; request=Send {} as JSON body; expected=Requiredness/stat us are unspecified; record observed behavior without mandatory 400 expectation.	VALID	Meaningful empty-object request-shape exploration with unspecified requiredness preserved. Evidence basis: exploratory; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: schema and request shape. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-040; title=Array instead of JSON object; preconditions=Valid user JWT; request=Send a JSON array body; expected=Wrong-shape robustness behavior is SPEC GAP.	VALID	Meaningful array-versus-object request-shape robustness exploration. Evidence basis: exploratory; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: schema and request shape. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-041; title=Unexpected additional property; preconditions=Valid user JWT; request=Send valid editable fields plus unexpected property; expected=Observe ignore/reject/persist behavior; no documented additional-property rule.	VALID	Meaningful additional-property exploration where the specification is silent. Evidence basis: exploratory; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: schema and request shape. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-042; title=Malformed JSON body; preconditions=Valid user JWT and JSON content type; request=Send malformed JSON; expected=Parser robustness behavior/status is SPEC GAP.	VALID	Meaningful malformed-JSON parser robustness exploration. Evidence basis: exploratory; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: robustness combinations. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-043; title=Shortest valid phone with Unicode text; preconditions=Valid user JWT; request=Use 10-digit valid phone with Unicode name/address; expected=Phone satisfies	INVALID	Combination of existing valid-phone and Unicode partitions without a distinct new risk. Evidence basis: official_contract_plus_exploration; requirement references: FR-04.	Rejected from the reviewed executable suite; covered by the existing valid-phone and Unicode cases.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
	documented valid boundary; inspect text persistence.			
Artifact context: Guided FR-04 generation: robustness combinations. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-044; title=Longest valid phone with empty text; preconditions=Valid user JWT; request=Use 11-digit valid phone with empty name/address; expected=Phone is valid; text requiredness is unknown and must be observed.	INVALID	Combination of existing 11-digit boundary and empty-text partitions without sufficient new value. Evidence basis: exploratory; requirement references: FR-04.	Rejected from the reviewed executable suite; covered by the existing 11-digit and empty-text cases.
Artifact context: Guided FR-04 generation: robustness combinations. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-045; title=Invalid phone with wrong-type text; preconditions=Valid user JWT; request=Use 9-digit phone with wrong-type name; expected=Negative robustness case; exact status is SPEC GAP.	INVALID	Combines two invalid dimensions and creates masking; duplicates existing single-fault tests. Evidence basis: exploratory; requirement references: FR-04.	Rejected from the reviewed executable suite; covered by existing single-fault phone and type cases.
Artifact context: Guided FR-04 generation: robustness combinations. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-046; title=Repeated identical update; preconditions=Valid user JWT; request=Send the same valid update twice, then GET profile; expected=Final state is stable and equals submitted values; exact response behavior is SPEC GAP.	VALID	Verifies repeated identical update behavior and final-state stability. Evidence basis: exploratory; requirement references: FR-04.	Accepted as-is; no correction required.
Artifact context: Guided FR-04 generation: robustness combinations. Tool: Codex. Exact standalone prompt text was not separately captured; this is the preserved artifact-level source context.	Verbatim registry artifact excerpt: ID=FR04-AI-047; title=Duplicate JSON phone keys; preconditions=Valid user JWT; request=Send body with duplicate phone keys; expected=Parser behavior is exploratory and must be observed, not assumed.	INVALID	Duplicate JSON keys have unstable parser semantics and no useful official oracle for this scoped API; low-value compared with meaningful required coverage. Evidence basis: exploratory; requirement references: FR-04.	Rejected from the reviewed executable suite; no replacement AI artifact is fabricated.

3A. AI-Assisted FR-04 Postman correction interaction

This addendum records the current correction interaction separately from the 47 original AI-generated test-case artifacts. It does not change their verdicts or the 21/16/10 summary. The exact user prompt is the complete current conversation turn beginning: "I reviewed the current FR-04 Postman artifacts and I do **not** approve official Newman execution yet." It is preserved as the actual user message in the session record and is not rewritten here. The exact interaction time was not separately captured; the interaction date is 22/08/2026. Tool: Codex.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user correction prompt: complete current conversation turn beginning "I reviewed the current FR-04 Postman artifacts and I do not approve official Newman execution yet." The prompt required correction of the existing FR-04 Postman Build, deterministic validation, no Newman execution, and an authorized audit update. Tool: Codex.	Verbatim artifact/result summary: corrected FR04-profile.postman_collection.json , added FR04-profile-stateful.postman_collection.json , corrected FR04-cases.postman_data.json to 41 rows, preserved exact null , added Unicode/additional-property stimuli, implemented real AI-031/AI-046/H-005	VALID	The correction output addresses the requested harness defects against the canonical test-case/review registries and verified FR-04 contract. Deterministic checks passed: 44 represented cases, 10 INVALID excluded, 7 HUMAN represented, canonical registries unchanged, Postman JSON/script preflight passed,	Student identified false-pass/false-failure risks, required the hybrid collection design, required no execution, and explicitly authorized this AI Audit Report update. Remaining manual/UI verification for SEC-04 and any runtime behavior are not claimed.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Date: 22/08/2026. Exact time: not separately captured.	sequences, strengthened read-back and cleanup checks, updated <code>submission-checklist.md</code> , and ran <code>JSON/script/canonical-count/derive/validate</code> preflights. No Newman run or runtime evidence was produced.		valid expired-token signature/exp preflight passed, and repository validation passed.	

The exact prompt time remains an unresolved audit field because it was not separately recorded by the conversation/runtime evidence. No earlier prompt or output is reconstructed in this addendum.

3B. AI-Assisted FR-04 Targeted Bug-Review Correction and Rerun

This addendum records the subsequent AI-assisted correction/rerun interaction separately from the 47 original AI-generated test-case artifacts. It does not change the original artifact verdicts or the 21/16/10 summary. The exact user prompt is the complete current conversation turn beginning: "Continue HW06 from the latest repository checkpoint." It is preserved in the session record and is not rewritten here. The exact interaction time was not separately captured; the interaction date is 23/08/2026. Tool: Codex.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user correction/rerun prompt: complete current conversation turn beginning "Continue HW06 from the latest repository checkpoint." The prompt required correction of H-001/H-002/H-003 mappings, targeted clean-state reruns, AI-027 isolation, AI-009 provenance verification, evidence registration, validation, HANDOFF update, no Issue publication, and no AI Audit Report update until explicitly authorized. Tool: Codex. Date: 23/08/2026. Exact time: not separately captured.	Verbatim evidence/result summary: corrected FR04-profile.postman_collection.json; H-001 sent JSON number 1234567890, H-002 sent JSON null, and H-003 sent shipping_address as an array; targeted Newman runs recorded clean-state phone and role reproductions; AI-027 was rerun independently; AI-009 was verified as a validly signed expired JWT and rerun; runs.json, bugs.json, and submission-checklist.md were updated; npm run hw06:derive and npm run hw06:validate passed with 0 errors and 0 warnings; no GitHub Issue was created. Evidence: provenance/run-index.md (RUN-20260823022303042-fr04-harness-corrected), RUN-20260823022316955-fr04-phone-corrected, RUN-20260823022329715-fr04-role-corrected, RUN-20260823022342040-fr04-ai027-corrected, and RUN-20260823022618165-fr04-ai009-provenance.	VALID	The output satisfies the requested targeted workflow and preserves the distinction between exploratory observations, verified contract violations, contaminated historical results, unresolved SPEC-GAP behavior, and later human bug confirmation. Evidence is attributable to real Newman reports and repository validation; no unsupported external publication is claimed.	The student reviewed the real Newman evidence, explicitly classified the affected cases, authorized the targeted correction/rerun phase, and later explicitly confirmed the phone-format and protected-field bugs after reviewing clean reproductions.

The exact prompt time remains an unresolved audit field because it was not separately recorded by the conversation/runtime evidence. At this interaction boundary, the two findings had not yet been published; their later human confirmation and publication are audited below.

3C. AI-Assisted FR-04 GitHub Issue artifacts

These rows audit the two public bug-report artifacts produced after the student reviewed the drafts, supplied conservative corrections, and explicitly authorized publication. Date: 23/08/2026. Tool: Codex. Exact chat interaction times were not separately captured; the canonical GitHub publication timestamps are retained in `provenance/bugs.json`.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user publication context: publish the confirmed FR-04 phone-format defect only after applying the supplied title and conservative specification-grounded corrections; use verified case IDs and committed Newman screenshots; do not invent status codes or public workflow text. Tool: Codex. Date: 23/08/2026. Exact time: not separately captured.	Verbatim final artifact identifiers: title= [HW06][FR-04] PUT /api/users/me persists phone values outside the documented format ; GitHub Issue #13 ; URL= https://github.com/BuhDuy256/automation-testing-hw04/issues/13 ; canonical cases= FR04-AI-013 , FR04-AI-014 , FR04-AI-015 , FR04-AI-016 , FR04-AI-017 , FR04-H-004 .	VALID	The published artifact is traceable to the FR-04 phone contract, canonical cases, clean targeted Newman evidence, human-confirmed bug record, and attested Issue-page evidence. It does not invent an HTTP status or error schema. Evidence: <code>provenance/bugs.json</code> , <code>provenance/evidence.json</code> , and <code>RUN-20260823022316955-fr04-phone-corrected</code> .	The student supplied the final title, required conservative wording, removed internal human-confirmation text, reviewed the resulting real Issue page, and explicitly attested <code>EVID-FR04-PHONE-ISSUE</code> .
Actual user publication context: publish the confirmed FR-04/SEC-06 protected-role defect as the sole primary finding; use <code>FR04-AI-026</code> and the clean role run; exclude email mutation and unsupported proof of admin-only action. Tool: Codex. Date: 23/08/2026. Exact time: not separately captured.	Verbatim final artifact identifiers: title= [HW06][FR-04][SEC-06] PUT /api/users/me allows a client to persist role=admin ; GitHub Issue #14 ; URL= https://github.com/BuhDuy256/automation-testing-hw04/issues/14 ; primary canonical case= <code>FR04-AI-026</code> .	VALID	The published artifact stays within the verified baseline role= <code>user</code> , submitted role= <code>admin</code> , and persisted role= <code>admin</code> evidence. It identifies a protected-role boundary defect without claiming execution of an admin-only operation. Evidence: <code>provenance/bugs.json</code> , <code>provenance/evidence.json</code> , and <code>RUN-20260823022329715-fr04-role-corrected</code> .	The student supplied the final title, restricted the public finding to role tampering, excluded the email observation, reviewed the real Issue page, and explicitly attested <code>EVID-FR04-ROLE-ISSUE</code> .

3D. AI-Assisted FR-04 CI workflow artifacts

These rows distinguish the stable-case CI behavior demonstration from complete canonical-suite integration. Date: 23/08/2026. Tool: Codex. Commit and GitHub Actions timestamps are canonical external provenance, not reconstructed chat times.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user CI context: implement reproducible GitHub Actions plus one all-pass run and one transparent intentional-single-failure run without weakening known-bug tests, deleting cases, changing the SUT, or fabricating evidence. Tool: Codex. Date: 23/08/2026. Exact time: not separately captured.	Verbatim artifact/result identifiers: workflow= https://github.com/BuhDuy256/automation-testing-hw04/blob/hw06-api-testing/.github/workflows/hw06-fr04-ci.yml ; stable case= <code>FR04-AI-001</code> ; all-pass run= 32617199599 ; intentional-failure run= 32617321761 ; intentional assertion= <code>FR04-AI-001 [CI-DEMO]</code> intentional single failure .	INCOMPLETE	The workflow and two real runs validly demonstrate CI mechanics, artifact preservation, one green result, and exactly one transparent deliberate failure. However, the sample executes only <code>FR04-AI-001</code> , so it does not prove the assignment phrase "all API test cases passing" for all 44 canonical FR-04 cases. Evidence: <code>provenance/ci-runs.json</code> and the two human-attested smoke screenshots.	The student explicitly challenged the one-case interpretation, required the report to disclose the conflict, prohibited a false <code>COMPLETE</code> status, and required a separate complete canonical-suite CI execution while preserving the smoke runs honestly.
Actual user correction context: add a separate GitHub Actions path that runs all canonical FR-04 execution groups with unchanged oracles, preserves reports/logs/header	Verbatim artifact/result identifiers: workflow= https://github.com/BuhDuy256/automation-testing-hw04/blob/hw06-api-testing/.github/workflows/hw06-fr04-canonical-	VALID	The workflow executes the complete canonical scope, restarts/reseeds the SUT for independent groups, preserves JSON/HTML/stdbackend artifacts, and leaves the job red when canonical tests fail.	The student required the separate full-suite design, inspected the real failed run screenshot, explicitly attested <code>EVID-CI-FR04-CANONICAL-FULL-SUITE</code> , and retained strict all-tests-green compliance as <code>PARTIAL</code>

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
evidence, exposes the real red result, and documents the strict-green limitation. Tool: Codex. Date: 23/08/2026. Exact time: not separately captured.	full-suite.yml ; commit= f34b71b24f368c819140fa077ce8942b76893470 ; run= 32618732832 ; result= failure ; logical counts= 44 executed, 35 PASS, 9 FAIL ; header coverage= 305/305 ; limitation= PARTIAL / documented limitation due to confirmed SUT defects .		CI results reconcile with local canonical results: 8 known-bug failures and 1 FR04-H-007 specification-gap observation. No oracle or SUT behavior was weakened.	rather than converting evidence attestation into false requirement completion.

3E. FR-08 test-case artifacts

These rows audit the 57 original AI-generated FR-08 test candidates for `POST /api/checkout` . Unlike the FR-04 phase, the exact prompts were captured prospectively: each candidate stores its verbatim bounded prompt, its batch id, the actual tool, and the real generation time, and the same prompts are held as files under `provenance/README.md` . Nothing in these rows is reconstructed from an output.

Tool for every row: Claude Opus 5 (claude-opus-5) via Claude Code. Generation batches and real timestamps: FR08-GEN-B1 18 candidates, FR08-GEN-B2 12 candidates, FR08-GEN-B3 10 candidates, FR08-GEN-B4 10 candidates, FR08-GEN-B5 6 candidates, FR08-GEN-B6 1 candidates. The Prompt + Tool cell carries a labelled verbatim excerpt of the stored prompt because the full bounded prompt runs to roughly two kilobytes per batch; the complete text is preserved verbatim in the repository at the path named in each cell.

Verdicts are the student's canonical review verdicts recorded in `provenance/human-reviews.json` at 2026-08-23T07:33:31.094Z. Each verdict judges whether the candidate fulfilled its own bounded prompt: the six INVALID rows failed the prompt's explicit duplication and closure constraints, and the eleven INCOMPLETE rows over-asserted or under-specified against the authoritative sources. The five student-selected extension cases FR08-H-001 through FR08-H-005 are not audited here because their final designs were selected and approved by the student rather than generated as AI artifacts; the AI-proposed shortlist that preceded them is recorded in `provenance/README.md` .

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1 , held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of <code>POST /api/checkout</code> , plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: <code>total_amount</code> partitions and boundaries, <code>shipping_address</code> partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client <code>total_amount</code> ." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	Verbatim registry artifact excerpt: ID=FR08-AI-001; title=Checkout with client total equal to the cart-derived total; request=POST <code>/api/checkout</code> with Authorization: Bearer <user JWT>, Content-Type: application/json, body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=Request is accepted for a logged-in user and an order is created whose persisted total equals the cart-derived total 200000, read back through GET <code>/api/orders/my-orders</code> ; the exact success status code and response schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in <code>provenance/test-cases.json</code> .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST /api/checkout, plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: total_amount partitions and boundaries, shipping_address partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount". Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	Verbatim registry artifact excerpt: ID=FR08-AI-002; title=Client total lower than the cart-derived total must not control the persisted total; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 1000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=The client value 1000 must not become the persisted order total; FR-08 requires the backend to recalculate the total from the cart, so either the request is rejected or the persisted total read back from GET /api/orders/my-orders equals 200000; the exact status and error schema are SPEC GAP.	VALID	Directly tests the explicit FR-08 rule that the backend recalculates the total and does not accept the client value; single injected fault with a persisted-state read-back. Evidence basis: official_contract; requirement references: FR-08; source anchors preserved in provenance/test-cases.json.	Accepted with one traceability correction: the security coverage tag is not supported by the requirement references, so treat this as a domain/cart-derived-total test in the reviewed usable view. The original AI candidate is preserved unchanged as generation provenance.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST /api/checkout, plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: total_amount partitions and boundaries, shipping_address partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount". Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	Verbatim registry artifact excerpt: ID=FR08-AI-003; title=Client total higher than the cart-derived total must not control the persisted total; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 9999000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=The inflated client value 9999000 must not become the persisted order total; the request is rejected or the persisted total read back equals the cart-derived 200000 per FR-08; the exact status and error schema are SPEC GAP.	VALID	Distinct inflation partition of the same explicit FR-08 rule, with a different real-world risk from the deflation case. Evidence basis: official_contract; requirement references: FR-08; source anchors preserved in provenance/test-cases.json.	Accepted with one traceability correction: the security coverage tag is not supported by the requirement references, so treat this as a domain/cart-derived-total test in the reviewed usable view. The original AI candidate is preserved unchanged as generation provenance.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST	Verbatim registry artifact excerpt: ID=FR08-AI-004; title=Zero client total with a non-empty cart; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 0, "shipping_address": "123 Le Loi, TP.HCM"}; expected=Zero is the lower boundary of the client-supplied amount partition; no	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract_plus_exploration; requirement	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
/api/checkout , plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: total_amount partitions and boundaries, shipping_address partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount "." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	status is documented for it, so the only asserted invariant is that a persisted order total of 0 must not result from a cart worth 200000 (FR-08); the exact status and error schema are SPEC GAP.		references: FR-08; source anchors preserved in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST /api/checkout , plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: total_amount partitions and boundaries, shipping_address partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount "." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	Verbatim registry artifact excerpt: ID=FR08-AI-005; title=Negative client total with a non-empty cart; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": -200000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=A negative client amount must not become the persisted order total; either the request is rejected or the persisted total equals the cart-derived 200000 per FR-08; the exact status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract_plus_exploration; requirement references: FR-08; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST /api/checkout , plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: total_amount partitions and boundaries, shipping_address partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount "." Tool:	Verbatim registry artifact excerpt: ID=FR08-AI-006; title=total_amount omitted from the request body; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"shipping_address": "123 Le Loi, TP.HCM"} with no total_amount key; expected=Requiredness of total_amount is not documented; observe the response without assuming a status, and assert only that if an order is created its persisted total equals the cart-derived 200000 per FR-08; the exact status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract_plus_exploration; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.				
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST /api/checkout, plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: total_amount partitions and boundaries, shipping_address partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount"." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	Verbatim registry artifact excerpt: ID=FR08-AI-007; title=total_amount explicitly null; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": null, "shipping_address": "123 Le Loi, TP.HCM"}; expected=Null handling is undocumented and distinct from omission; observe the response and assert only that any created order carries the cart-derived total 200000 per FR-08; the exact status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json.	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST /api/checkout, plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: total_amount partitions and boundaries, shipping_address partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount"." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	Verbatim registry artifact excerpt: ID=FR08-AI-008; title=total_amount sent as a numeric string; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": "200000", "shipping_address": "123 Le Loi, TP.HCM"}; expected=The specification shows total_amount as a number but defines no type-validation behavior; observe the response and assert only that any persisted total equals the cart-derived 200000 per FR-08; the exact status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json.	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required	Verbatim registry artifact excerpt: ID=FR08-AI-009; title=total_amount sent as a non-numeric string; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": "abc",	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of <code>POST</code> <code>/api/checkout</code> , plus the documented cart-derived- total invariant. - Parameters/states/SEC/sche ma rules in scope: <code>total_amount</code> partitions and boundaries, <code>shipping_address</code> partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client <code>total_amount</code> "." Tool: Claude Opus 5 (claude- opus-5) via Claude Code. Generated at: 2026-08- 23T06:51:51.638Z.	"shipping_address": "123 Le Loi, TP.HCM"); expected=Type-validation behavior is undocumented; observe the response and assert only that no order is persisted with a non-numeric or corrupted total and that any created order carries the cart-derived 200000 per FR- 08; the exact status and error schema are SPEC GAP.		duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR- 08, API-SPEC-4.3; source anchors preserved in <code>provenance/test- cases.json</code> .	
Verbatim excerpt of the stored bounded prompt for batch <code>FR08-GEN-B1</code> , held in full at <code>provenance/README.md</code> and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of <code>POST</code> <code>/api/checkout</code> , plus the documented cart-derived- total invariant. - Parameters/states/SEC/sche ma rules in scope: <code>total_amount</code> partitions and boundaries, <code>shipping_address</code> partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client <code>total_amount</code> "." Tool: Claude Opus 5 (claude- opus-5) via Claude Code. Generated at: 2026-08- 23T06:51:51.638Z.	Verbatim registry artifact excerpt: ID=FR08-AI-010; title=Fractional client total just below the cart-derived total; request=POST <code>/api/checkout</code> with Authorization: Bearer <user JWT> and body { <code>"total_amount"</code> : 199999.99, <code>"shipping_address"</code> : "123 Le Loi, TP.HCM"}; expected=A fractional value just below the cart total is the nearest- neighbour boundary of the mismatch partition; the persisted total must still be the cart-derived 200000 per FR-08 rather than 199999.99; the exact status and error schema are SPEC GAP.	INCOMPLETE	The candidate frames itself as the nearest-neighbour boundary of the mismatch partition, which restates the purpose of FR08-AI-002. Its genuinely distinct risk is that the client submits a fractional forged total. Evidence basis: official_contract_plus_explor ation; requirement references: FR-08; source anchors preserved in <code>provenance/test- cases.json</code> .	Keep the concrete inputs cart-derived total 200000 and client <code>total_amount</code> 199999.99. Corrected oracle: the backend derives the authoritative total from cart state, the client-supplied 199999.99 must not control the persisted order total, and the resulting authoritative total remains the cart-derived value. Do not assert that fractional values are forbidden or that persisted totals must be integers; no authoritative source states that. Exact undocumented HTTP/error behavior remains SPEC GAP.
Verbatim excerpt of the stored bounded prompt for batch <code>FR08-GEN-B1</code> , held in full at <code>provenance/README.md</code> and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of <code>POST</code> <code>/api/checkout</code> , plus the documented cart-derived- total invariant. - Parameters/states/SEC/sche ma rules in scope: <code>total_amount</code> partitions and boundaries, <code>shipping_address</code>	Verbatim registry artifact excerpt: ID=FR08-AI-011; title=Extremely large client total; request=POST <code>/api/checkout</code> with Authorization: Bearer <user JWT> and body { <code>"total_amount"</code> : 999999999999999999, <code>"shipping_address"</code> : "123 Le Loi, TP.HCM"}; expected=The upper magnitude boundary of the client amount partition is undocumented; assert only that the persisted total equals the cart-derived 200000 per FR-08 and that the endpoint returns a structured HTTP response rather than failing without	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API- specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR- 08; source anchors preserved in <code>provenance/test- cases.json</code> .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount "." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	one; the exact status and any numeric ceiling are SPEC GAP.			
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST /api/checkout, plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: total_amount partitions and boundaries, shipping_address partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount "." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	Verbatim registry artifact excerpt: ID=FR08-AI-012; title=Shipping address containing Vietnamese diacritics; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "12 Duong Nguyen Van Cu, Phuong 4, Quan 5, Thanh pho Ho Chi Minh"} sent with UTF-8 Vietnamese diacritics on every accented syllable; expected=A Vietnamese-language address is inside the valid string partition; the order is created and the address read back through GET /api/orders/:id is character-identical to the sent value with no encoding corruption; the exact success status and response schema are SPEC GAP.	INCOMPLETE	The candidate states that the address carries Vietnamese diacritics but the literal value recorded in the request has none, so it is not executable exactly as written. Evidence basis: official_contract_plus_exploration; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json.	Use the exact literal value 12 Duong Nguyen Van Cu, Phuong 4, Quan 5, Thanh pho Ho Chi Minh written with real Vietnamese Unicode diacritics on every accented syllable. Verify exact persisted read-back text where the authoritative observation mechanism supports it, and record an evidence gap if the order-detail endpoint does not return the stored address.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST /api/checkout, plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: total_amount partitions and boundaries, shipping_address partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount "." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	Verbatim registry artifact excerpt: ID=FR08-AI-013; title=Empty-string shipping address; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": ""}; expected=No validation rule for an empty address is documented; observe the response without assuming rejection and record whether an order is persisted without a delivery address; the exact status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json.	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1, held in	Verbatim registry artifact excerpt: ID=FR08-AI-014; title=Whitespace-only	VALID	Executable as written from a stated starting state, one injected fault, and an oracle	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST /api/checkout , plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: total_amount partitions and boundaries, shipping_address partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount "." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	shipping address; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": " "} containing exactly three space characters; expected=Trimming and blank-address behavior is undocumented; observe the response and record whether the stored address is trimmed, stored as blank, or rejected; the exact status and error schema are SPEC GAP.		anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST /api/checkout , plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: total_amount partitions and boundaries, shipping_address partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client total_amount "." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	Verbatim registry artifact excerpt: ID=FR08-AI-015; title=Explicitly null shipping address; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": null}; expected=Nullability of shipping_address is not documented; observe the response and record whether a null address is persisted; the exact status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of POST /api/checkout , plus the documented cart-derived-total invariant. -	Verbatim registry artifact excerpt: ID=FR08-AI-016; title=shipping_address omitted from the request body; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000} with no shipping_address key; expected=Requiredness of shipping_address is not documented; observe the response and record whether an order is persisted with no address at all; the exact	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract_plus_exploration; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Parameters/states/SEC/schema rules in scope: <code>total_amount</code> partitions and boundaries, <code>shipping_address</code> partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client <code>total_amount</code> ". Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	status and error schema are SPEC GAP.		<code>provenance/test-cases.json</code> .	
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1, held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of <code>POST /api/checkout</code> , plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: <code>total_amount</code> partitions and boundaries, <code>shipping_address</code> partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client <code>total_amount</code> ". Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:51:51.638Z.	Verbatim registry artifact excerpt: ID=FR08-AI-017; title= <code>shipping_address</code> sent with a numeric type; request= <code>POST /api/checkout</code> with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": 12345}; expected=The specification shows <code>shipping_address</code> as text but defines no type validation; observe the response and record whether a non-string address is coerced, stored, or rejected; the exact status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in <code>provenance/test-cases.json</code> .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B1, held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: equivalence partitioning and boundary value analysis on the two documented body fields of <code>POST /api/checkout</code> , plus the documented cart-derived-total invariant. - Parameters/states/SEC/schema rules in scope: <code>total_amount</code> partitions and boundaries, <code>shipping_address</code> partitions and boundaries, FR-08 rule "backend recalculates the total and does not accept the client <code>total_amount</code> ". Tool: Claude Opus 5 (claude-opus-5) via Claude Code.	Verbatim registry artifact excerpt: ID=FR08-AI-018; title=Shipping address of 1000 characters; request= <code>POST /api/checkout</code> with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "<the ASCII character A repeated exactly 1000 times>"}; expected=No maximum length is documented for <code>shipping_address</code> ; observe the response and, if an order is created, assert that <code>GET /api/orders/:id</code> returns the address at its full 1000-character length without silent truncation; the exact status and any length limit are SPEC GAP.	INCOMPLETE	The specification defines no maximum length and no preservation or truncation rule for <code>shipping_address</code> , but the candidate turns the exploratory 1000-character input into a requirement that the stored value must remain exactly 1000 characters. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in <code>provenance/test-cases.json</code> .	Keep the concrete 1000-character input as a robustness partition. Corrected oracle: observe whether the request is accepted or rejected; if an order is created, record the returned address length and value through the available read-back; do not classify truncation, normalization, or rejection as a contract violation unless an authoritative source supports that judgment; any created order must still respect the documented cart-derived-total invariant. Exact length and normalization behavior remains SPEC GAP.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Generated at: 2026-08-23T06:51:51.638Z.				
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/schema rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	Verbatim registry artifact excerpt: ID=FR08-AI-019; title=Checkout without an Authorization header; request=POST /api/checkout with no Authorization header and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=FR-08 restricts checkout to logged-in users and SEC-02 requires a valid JWT on secured APIs, so the request must be refused and no order may appear in GET /api/orders/my-orders for the cart owner; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: <code>official_contract</code>; requirement references: FR-08, SEC-02, API-SPEC-4; source anchors preserved in <code>provenance/test-cases.json</code> .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/schema rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	Verbatim registry artifact excerpt: ID=FR08-AI-020; title=Checkout with a syntactically malformed Authorization header; request=POST /api/checkout with header Authorization: Token abc123 instead of the Bearer scheme, and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=SEC-02 and the documented Bearer header rule require a valid JWT presented in the Bearer scheme, so a non-Bearer scheme must be refused and no order may be created; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: <code>official_contract</code>; requirement references: SEC-02, API-SPEC-4; source anchors preserved in <code>provenance/test-cases.json</code> .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at <code>provenance/README.md</code> and in this candidate's	Verbatim registry artifact excerpt: ID=FR08-AI-021; title=Checkout with a Bearer prefix but an empty token; request=POST /api/checkout with header Authorization:	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
generationPrompt field: "Required technique/category: authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/schema rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	Bearer (the scheme keyword followed by a single space and no token) and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=An empty credential is not a valid JWT, so SEC-02 requires refusal and no order may be created; the exact rejection status and error schema are SPEC GAP.		undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract; requirement references: SEC-02, API-SPEC-4; source anchors preserved in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/schema rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	Verbatim registry artifact excerpt: ID=FR08-AI-022; title=Checkout with a JWT whose signature has been tampered with; request=POST /api/checkout with header Authorization: Bearer <valid header and payload segments plus a signature whose last character is replaced by X> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=SEC-02 requires a valid token, so a token failing signature verification must be refused and no order may be created; the exact rejection status and error schema are SPEC GAP.	INCOMPLETE	The mutation rule replace the last character with X is not deterministic when the signature already ends in X, so two runs could send different tokens. Evidence basis: official_contract; requirement references: SEC-02; source anchors preserved in provenance/test-cases.json .	Make the tampering deterministic: replace the final signature character with A, or with B when it is already A. A fixed tampered token stored as an environment constant is equally acceptable and is preferred when byte-stability across runs matters.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/sche	Verbatim registry artifact excerpt: ID=FR08-AI-023; title=Checkout with a structurally invalid token string; request=POST /api/checkout with header Authorization: Bearer not-a-jwt and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=A string that is not a three-segment JWT cannot be a valid token, so SEC-02 requires refusal with a structured HTTP response	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract; requirement references: SEC-02; source anchors preserved in	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
ma rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	rather than an unhandled failure, and no order may be created; the exact rejection status and error schema are SPEC GAP.		provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/schema rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	Verbatim registry artifact excerpt: ID=FR08-AI-024; title=Checkout with a JWT signed by a different secret; request=POST /api/checkout with header Authorization: Bearer <HS256 JWT carrying the real user id and email but signed with wrong-secret-123> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=SEC-02 requires token validity to be verified rather than merely parsed, so a foreign-signed token must be refused and no order may be created for that user; the exact rejection status and error schema are SPEC GAP.	INVALID	Duplicate diagnostic of FR08-AI-022: a token signed with a foreign secret and a token whose signature was tampered with both fail at the same signature-verification step and assert the same invariant, while this one additionally requires an offline HMAC-SHA256 construction. Evidence basis: official_contract; requirement references: SEC-02; source anchors preserved in provenance/test-cases.json .	Drop FR08-AI-024. FR08-AI-022 remains the canonical invalid-signature diagnostic for the reviewed suite.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/schema rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across	Verbatim registry artifact excerpt: ID=FR08-AI-025; title=Order attribution: an order created by one user must not appear in another user history; request=POST /api/checkout with Authorization: Bearer <user A JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}; followed by GET /api/orders/my-orders with Authorization: Bearer <user B JWT>; expected=The order belongs to the authenticated caller, so it must appear in user A history and must not appear in user B history; SEC-02 ties the operation to the token identity; the exact response schema of the	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract_plus_exploration; requirement references: FR-08, SEC-02, API-SPEC-4.4; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	history endpoint is SPEC GAP.			
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/schema rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	Verbatim registry artifact excerpt: ID=FR08-AI-026; title=Client-supplied user_id must not redirect order ownership; request=POST /api/checkout with Authorization: Bearer <user A JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM", "user_id": <user B numeric id>} where user_id is an extra field not present in the documented request body; expected=Order ownership is derived from the authenticated token, not from the request body, so the created order must not be attributed to user B; read back both histories and assert user B count is unchanged; the exact status and whether unknown fields are rejected are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, SEC-02, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/schema rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	Verbatim registry artifact excerpt: ID=FR08-AI-027; title=Client-supplied order status must not control server-owned order state; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM", "status": "delivered"} where status is an extra field not present in the documented request body; expected=The documented request body does not include status, and order status transitions are described as a server-side concern with cancellation handled by its own endpoint; read the created order through GET /api/orders/:id and record whether the client value was persisted; the exact required status valu ... (truncated verbatim excerpt).	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3, API-SPEC-4.6; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/schema rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	Verbatim registry artifact excerpt: ID=FR08-AI-028; title=HTML payload in the shipping address is stored and returned unchanged by the API; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "<script>alert('xss')</script>123 Le Loi"}; expected=SEC-04 places the escaping obligation at the UI display boundary, not at the API, so the API-level oracle is only that the request is handled without an unhandled failure and that GET /api/orders/:id returns the stored value; whether the payload is rejected at the API is SPEC GAP, and this case cann ... (truncated verbatim excerpt).	INCOMPLETE	The oracle is internally inconsistent: the title asserts the value is stored and returned unchanged while the body records API-side rejection as SPEC GAP. Listing SEC-04 in the requirement references also reads as a SEC-04 conformance claim that API read-back cannot support. Evidence basis: exploratory; requirement references: SEC-04, FR-08; source anchors preserved in provenance/test-cases.json .	Limit the purpose explicitly to preserving the HTML-like value through the API while recording that UI and display safety requires separate evidence. Make the oracle conditional: if the request is accepted, the read-back value is character-identical; if it is rejected, no order is created. State in the case that API-side observation is not UI/display-boundary SEC-04 verification, so SEC-04 conformance is never claimed from this case.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/schema rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	Verbatim registry artifact excerpt: ID=FR08-AI-029; title=SQL metacharacter payload in the shipping address must not damage order persistence; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "1 Le Loi"}; DROP TABLE orders; -- "; expected=SEC-05 requires parameterized queries; the observable black-box oracle is that the endpoint returns a structured HTTP response and that GET /api/orders/my-orders still works afterwards with the previously created orders intact, so no schema damage occurred; a safe response does not prove parameteriz ... (truncated verbatim excerpt).	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: SEC-05, FR-08; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B2 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category:	Verbatim registry artifact excerpt: ID=FR08-AI-030; title=SQL metacharacter payload in total_amount must not alter the persisted total or damage persistence; request=POST /api/checkout with Authorization: Bearer <user JWT> and body	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
authentication boundary, ownership/attribution, client-controlled protected fields, and source-defined security rules. - Parameters/states/SEC/schema rules in scope: SEC-02 valid-JWT requirement, SEC-04 display-boundary handling of the user-supplied address, SEC-05 parameterized-query expectation around order persistence, order ownership/attribution across two users, and extra client-supplied body fields that must no ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:53:35.613Z.	{"total_amount": "1 OR 1=1", "shipping_address": "123 Le Loi, TP.HCM"}; expected=Per FR-08 the persisted total must come from the cart, so any created order must carry 200000, and per SEC-05 the endpoint must return a structured HTTP response with GET /api/orders/my-orders still functioning afterwards; the exact status is SPEC GAP and the result cannot prove the internal query t ... (truncated verbatim excerpt).		reviewed case. Evidence basis: exploratory; requirement references: SEC-05, FR-08; source anchors preserved in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B3 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: state-transition testing with an explicit starting state, action, expected next state, authoritative read-back, and reset plan. - Parameters/states/SEC/schema rules in scope: cart cleared after a successful checkout, order created and visible in the personal order history, the total derived from real cart contents including multi-item and aggregated-quantity carts, empty-cart checkout, repeated checkout, and cart iso ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:55:04.569Z.	Verbatim registry artifact excerpt: ID=FR08-AI-031; title=Successful checkout clears the cart; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by GET /api/cart with the same token; expected=FR-08 states that the cart is cleared after a successful checkout, so the post-action GET /api/cart must return no cart lines for that user; the exact empty-cart response shape is SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract; requirement references: FR-08, API-SPEC-4.1; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B3 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: state-transition testing with an explicit starting state, action, expected next state, authoritative read-back, and reset plan. - Parameters/states/SEC/schema rules in scope: cart cleared after a successful checkout, order created and visible in the personal order history, the total derived from	Verbatim registry artifact excerpt: ID=FR08-AI-032; title=Successful checkout creates exactly one order in the personal order history; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by GET /api/orders/my-orders with the same token; expected=A successful checkout is an order-creating transition, so the order count must become exactly N+1 and the newest entry must be the order just created; the	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract_plus_exploration; requirement references: FR-08, API-SPEC-4.4; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
real cart contents including multi-item and aggregated-quantity carts, empty-cart checkout, repeated checkout, and cart iso ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:55:04.569Z.	exact order object schema is SPEC GAP.			
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B3, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: state-transition testing with an explicit starting state, action, expected next state, authoritative read-back, and reset plan. - Parameters/states/SEC/schema rules in scope: cart cleared after a successful checkout, order created and visible in the personal order history, the total derived from real cart contents including multi-item and aggregated-quantity carts, empty-cart checkout, repeated checkout, and cart iso ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:55:04.569Z.	Verbatim registry artifact excerpt: ID=FR08-AI-033; title=Persisted total for a two-product cart equals the sum of the line amounts; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 350000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by GET /api/orders/:id for the created order; expected=FR-08 requires the total to be calculated from the cart, so the persisted order total must equal 100000*2 + 50000*3 = 350000; the exact response schema is SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract; requirement references: FR-08, API-SPEC-4.5; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B3, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: state-transition testing with an explicit starting state, action, expected next state, authoritative read-back, and reset plan. - Parameters/states/SEC/schema rules in scope: cart cleared after a successful checkout, order created and visible in the personal order history, the total derived from real cart contents including multi-item and aggregated-quantity carts, empty-cart checkout, repeated checkout, and cart iso ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:55:04.569Z.	Verbatim registry artifact excerpt: ID=FR08-AI-034; title=Persisted total reflects the aggregated quantity when the same product is added twice; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 500000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by GET /api/orders/:id for the created order; expected=FR-07 requires re-adding a product to increase its quantity rather than create a new line, and FR-08 requires the total to come from the cart, so the persisted total must be 500000; the exact response schema is SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract; requirement references: FR-07, FR-08, API-SPEC-4.2; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for	Verbatim registry artifact excerpt: ID=FR08-AI-035;	INCOMPLETE	The setup is not reliably executable: using a session	Reach the empty-cart state deterministically by

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
batch FR08-GEN-B3 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: state-transition testing with an explicit starting state, action, expected next state, authoritative read-back, and reset plan. - Parameters/states/SEC/schema rules in scope: cart cleared after a successful checkout, order created and visible in the personal order history, the total derived from real cart contents including multi-item and aggregated-quantity carts, empty-cart checkout, repeated checkout, and cart iso ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:55:04.569Z.	title=Checkout with an empty cart; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by GET /api/orders/my-orders; expected=No status is documented for an empty-cart checkout, so the only asserted invariant is that a cart worth nothing must not produce an order carrying the client value 200000; record whether the order count stays at N or an order with total 0 is created; the exact status and error schema are SPEC GAP.		that has not added items does not empty an existing cart, because implementation cart state is process memory keyed to the user rather than to the login session. Evidence basis: official_contract_plus_exploration; requirement references: FR-08; source anchors preserved in provenance/test-cases.json .	registering a fresh user with POST /api/register, logging in, and checking out without ever calling POST /api/cart; restarting the backend immediately before the case is the fallback. Do not claim that logging in again clears the in-memory cart. The empty-cart outcome itself remains exploratory because the specification does not define it.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B3 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: state-transition testing with an explicit starting state, action, expected next state, authoritative read-back, and reset plan. - Parameters/states/SEC/schema rules in scope: cart cleared after a successful checkout, order created and visible in the personal order history, the total derived from real cart contents including multi-item and aggregated-quantity carts, empty-cart checkout, repeated checkout, and cart iso ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:55:04.569Z.	Verbatim registry artifact excerpt: ID=FR08-AI-036; title=Second checkout immediately after a successful checkout; request=POST /api/checkout a second time with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by GET /api/orders/my-orders; expected=Because FR-08 requires the cart to be cleared after success, the second checkout starts from an empty cart and must not create a second order carrying the client value 200000; record whether the count stays at N+1; the exact status and error schema are SPEC GAP.	VALID	The case does not need to invent the exact empty-cart response. After the first successful checkout FR-08 requires the cart to be cleared, so the second checkout must not use the client-supplied 200000 as an authoritative total for the now-empty cart. Evidence basis: official_contract_plus_exploration; requirement references: FR-08; source anchors preserved in provenance/test-cases.json .	Accepted as-is. The observed order count and the exact second-request behavior may remain exploratory and SPEC GAP.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B3 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: state-transition testing with an explicit starting state, action, expected next state, authoritative read-back, and reset plan. - Parameters/states/SEC/schema rules in scope: cart cleared after a successful checkout, order created and visible in the personal order history, the total derived from real cart contents including multi-item and aggregated-quantity carts, empty-cart checkout, repeated checkout, and cart iso ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:55:04.569Z.	Verbatim registry artifact excerpt: ID=FR08-AI-037; title=A rejected unauthenticated checkout must not modify the cart; request=POST /api/checkout with no Authorization header and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by GET /api/cart with the user token; expected=Cart clearing is documented only as a consequence of a successful	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract_plus_exploration; requirement references: FR-08, SEC-02, API-SPEC-4.1; source	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
ma rules in scope: cart cleared after a successful checkout, order created and visible in the personal order history, the total derived from real cart contents including multi-item and aggregated-quantity carts, empty-cart checkout, repeated checkout, and cart iso ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:55:04.569Z.	checkout, so after a refused unauthenticated attempt the cart must still contain product id 1 with quantity 2; the exact rejection status is SPEC GAP.		anchors preserved in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B3 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: state-transition testing with an explicit starting state, action, expected next state, authoritative read-back, and reset plan. - Parameters/states/SEC/schema rules in scope: cart cleared after a successful checkout, order created and visible in the personal order history, the total derived from real cart contents including multi-item and aggregated-quantity carts, empty-cart checkout, repeated checkout, and cart iso ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:55:04.569Z.	Verbatim registry artifact excerpt: ID=FR08-AI-038; title=Created order is retrievable by its own detail endpoint; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by GET /api/orders/:id using the captured identifier and the same token; expected=The checkout transition creates a persisted order, so the detail endpoint must return that same order with the cart-derived total 200000 and the submitted shipping address; the exact detail response schema is SPEC GAP, and if checkout returns no identifier the traceability failure is recorded as an ... (truncated verbatim excerpt).	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract_plus_exploration; requirement references: FR-08, API-SPEC-4.5; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B3 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: state-transition testing with an explicit starting state, action, expected next state, authoritative read-back, and reset plan. - Parameters/states/SEC/schema rules in scope: cart cleared after a successful checkout, order created and visible in the personal order history, the total derived from real cart contents including multi-item and aggregated-quantity carts, empty-cart checkout, repeated checkout, and cart iso ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code.	Verbatim registry artifact excerpt: ID=FR08-AI-039; title=One user checkout must not clear another user cart; request=POST /api/checkout with Authorization: Bearer <user A JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by GET /api/cart with Authorization: Bearer <user B JWT>; expected=Cart clearing is scoped to the checking-out user, so user B cart must still contain product id 1 with quantity 2 after user A checkout; the exact cart response shape is SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract_plus_exploration; requirement references: FR-08, SEC-02, API-SPEC-4.1; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Generated at: 2026-08-23T06:55:04.569Z.				
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B3 , held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: state-transition testing with an explicit starting state, action, expected next state, authoritative read-back, and reset plan. - Parameters/states/SEC/schema rules in scope: cart cleared after a successful checkout, order created and visible in the personal order history, the total derived from real cart contents including multi-item and aggregated-quantity carts, empty-cart checkout, repeated checkout, and cart iso ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:55:04.569Z.	Verbatim registry artifact excerpt: ID=FR08-AI-040; title=Cart is usable again after a successful checkout clears it; request=POST /api/cart {"id": 2, "name": "San pham B", "price": 50000, "quantity": 1} with the user token, followed by GET /api/cart, then POST /api/checkout with body {"total_amount": 50000, "shipping_address": "123 Le Loi, TP.HCM"} and GET /api/orders/:id for the new order; expected=After the post-checkout clearing required by FR-08, a new cart must build from an empty state, so the second order total must be exactly 50000 and must not include any residue of the earlier 200000 cart; the exact response schemas are SPEC GAP.	INCOMPLETE	The purpose is genuinely distinct from FR08-AI-031, because a cart can read as empty while the checkout derivation still uses stale state, but the case is framed as cart usability in an FR-07 flavour and never states how no residue is observed. Evidence basis: <code>official_contract_plus_exploration</code>; requirement references: FR-08, API-SPEC-4.2; source anchors preserved in <code>provenance/test-cases.json</code>.	Rewrite around the checkout-specific state transition: successful checkout, cart becomes empty, adding a new item establishes a new post-checkout cart, and the second order's server-derived total is exactly 50000 while purchased items do not reappear in it. State the observation explicitly: compare the second order's total through GET /api/orders/:id, and assert its item composition where the detail endpoint exposes items, otherwise record an evidence gap.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B4 , held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: request envelope and body-shape faults, plus response-shape observation where the specification defines no exact schema. - Parameters/states/SEC/schema rules in scope: absent body, empty JSON object, malformed JSON, wrong top-level JSON type, structured wrong type for a documented field, unknown additional properties, request Content-Type handling, and observation of the actual success response shape." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:56:49.430Z.	Verbatim registry artifact excerpt: ID=FR08-AI-041; title=Checkout request sent with no body at all; request=POST /api/checkout with Authorization: Bearer <user JWT>, header Content-Type: application/json, and no request body bytes; expected=Body requiredness is not documented, so the case observes the response and asserts only that the endpoint returns a structured HTTP response and that any created order carries the cart-derived total 200000 per FR-08; the exact status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in <code>provenance/test-cases.json</code>.	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B4 , held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: request envelope and body-shape faults, plus response-shape	Verbatim registry artifact excerpt: ID=FR08-AI-042; title=Checkout request with an empty JSON object body; request=POST /api/checkout with Authorization: Bearer <user JWT>, Content-Type: application/json, and body {}; expected=Both documented fields are absent at once; no validation contract is	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory;	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
observation where the specification defines no exact schema. - Parameters/states/SEC/schema rules in scope: absent body, empty JSON object, malformed JSON, wrong top-level JSON type, structured wrong type for a documented field, unknown additional properties, request Content-Type handling, and observation of the actual success response shape." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:56:49.430Z.	documented, so the case records the response and asserts only that any created order carries the cart-derived total 200000 per FR-08 and no address other than one the server itself supplies; the exact status and error schema are SP ... (truncated verbatim excerpt).		requirement references: FR-08, API-SPEC-4.3; source anchors preserved in <code>provenance/test-cases.json</code>.	
Verbatim excerpt of the stored bounded prompt for batch <code>FR08-GEN-B4</code>, held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: request envelope and body-shape faults, plus response-shape observation where the specification defines no exact schema. - Parameters/states/SEC/schema rules in scope: absent body, empty JSON object, malformed JSON, wrong top-level JSON type, structured wrong type for a documented field, unknown additional properties, request Content-Type handling, and observation of the actual success response shape." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:56:49.430Z.	Verbatim registry artifact excerpt: ID=FR08-AI-043; title=Checkout request with syntactically malformed JSON; request=POST /api/checkout with Authorization: Bearer <user JWT>, Content-Type: application/json, and the raw body <code>{"total_amount": 200000, "shipping_address":</code> with the closing brace and value omitted so the payload is not valid JSON; expected=Parser-failure behavior is not documented; the case asserts only that the endpoint returns a structured HTTP response rather than failing without one, and that no order is created from an unparseable payload; the exact status and error schema are SPEC GAP.	INCOMPLETE	The malformed payload is described narratively rather than as exact bytes, so two testers could send different inputs. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in <code>provenance/test-cases.json</code>.	Send exactly the raw body <code>{"total_amount": 200000, "shipping_address":</code> with no closing quote or brace, and require the Postman implementation to send that raw string without re-serializing it into valid JSON.
Verbatim excerpt of the stored bounded prompt for batch <code>FR08-GEN-B4</code>, held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: request envelope and body-shape faults, plus response-shape observation where the specification defines no exact schema. - Parameters/states/SEC/schema rules in scope: absent body, empty JSON object, malformed JSON, wrong top-level JSON type, structured wrong type for a documented field, unknown additional properties, request Content-Type handling, and observation of the actual success response shape." Tool: Claude Opus 5 (claude-	Verbatim registry artifact excerpt: ID=FR08-AI-044; title=Checkout request whose top-level JSON value is an array; request=POST /api/checkout with Authorization: Bearer <user JWT>, Content-Type: application/json, and body <code>[{"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}]</code>; expected=The documented request body is a JSON object; a top-level array is a structural mismatch with no documented handling, so the case asserts only a structured HTTP response and that any created order carries the cart-derived total 200000 per FR-08; the exact status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in <code>provenance/test-cases.json</code>.	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
opus-5) via Claude Code. Generated at: 2026-08-23T06:56:49.430Z.				
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B4 , held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: request envelope and body-shape faults, plus response-shape observation where the specification defines no exact schema. - Parameters/states/SEC/schema rules in scope: absent body, empty JSON object, malformed JSON, wrong top-level JSON type, structured wrong type for a documented field, unknown additional properties, request Content-Type handling, and observation of the actual success response shape." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:56:49.430Z.	Verbatim registry artifact excerpt: ID=FR08-AI-045; title=shipping_address sent as a nested JSON object; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": {"street": "123 Le Loi", "city": "TP.HCM"}}; expected=The specification shows shipping_address as text and defines no structured-object handling; the case records whether the object is rejected, stringified, or stored, using GET /api/orders/:id as the read-back; the exact status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in <code>provenance/test-cases.json</code> .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B4 , held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: request envelope and body-shape faults, plus response-shape observation where the specification defines no exact schema. - Parameters/states/SEC/schema rules in scope: absent body, empty JSON object, malformed JSON, wrong top-level JSON type, structured wrong type for a documented field, unknown additional properties, request Content-Type handling, and observation of the actual success response shape." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:56:49.430Z.	Verbatim registry artifact excerpt: ID=FR08-AI-046; title=total_amount sent as a boolean; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": true, "shipping_address": "123 Le Loi, TP.HCM"}; expected=A boolean is neither a number nor a numerically parseable string, and no type contract is documented; the case asserts only that any persisted total equals the cart-derived 200000 per FR-08 and that no boolean or coerced 1 is stored as the order total; the exact status and error schema are SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in <code>provenance/test-cases.json</code> .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B4 , held in full at <code>provenance/README.md</code> and in this candidate's <code>generationPrompt</code> field: "Required technique/category: request envelope and body-shape faults, plus response-shape	Verbatim registry artifact excerpt: ID=FR08-AI-047; title=Unknown additional property in the request body; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM", "discount_code":	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: exploratory;	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
observation where the specification defines no exact schema. - Parameters/states/SEC/schema rules in scope: absent body, empty JSON object, malformed JSON, wrong top-level JSON type, structured wrong type for a documented field, unknown additional properties, request Content-Type handling, and observation of the actual success response shape." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:56:49.430Z.	"FREESHIP99"; expected=No additional-property rule is documented, so the case records whether the unknown field is ignored or rejected and asserts only that it does not change the cart-derived persisted total of 200000 per FR-08; the exact status and any additional-property policy are SPEC GAP.		requirement references: FR-08, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B4 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: request envelope and body-shape faults, plus response-shape observation where the specification defines no exact schema. - Parameters/states/SEC/schema rules in scope: absent body, empty JSON object, malformed JSON, wrong top-level JSON type, structured wrong type for a documented field, unknown additional properties, request Content-Type handling, and observation of the actual success response shape." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:56:49.430Z.	Verbatim registry artifact excerpt: ID=FR08-AI-048; title=Checkout body sent with a text/plain Content-Type; request=POST /api/checkout with Authorization: Bearer <user JWT>, header Content-Type: text/plain, and the raw body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=No Content-Type requirement is documented for this endpoint; the case records whether the payload is parsed or ignored and asserts only that any created order carries the cart-derived total 200000 per FR-08 rather than an empty or default address; the exact status and content-type policy are SPEC GA ... (truncated verbatim excerpt).	INCOMPLETE	The oracle mixes unrelated fields by requiring that any created order carries 200000 rather than a default or empty address, which compares a numeric total against an address state and is not logically valid. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json .	Keep the purpose Content-Type text/plain with a JSON-looking payload. Corrected oracle: observe whether the endpoint parses or rejects the request, with exact Content-Type handling remaining SPEC GAP; if an order is created, its total must still be derived from the current cart; separately record the persisted shipping_address value if it is observable. Do not compare a numeric total against an address state.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B4 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: request envelope and body-shape faults, plus response-shape observation where the specification defines no exact schema. - Parameters/states/SEC/schema rules in scope: absent body, empty JSON object, malformed JSON, wrong top-level JSON type, structured wrong type for a documented field, unknown additional properties, request Content-Type handling, and observation of the actual success response shape." Tool: Claude Opus 5 (claude-	Verbatim registry artifact excerpt: ID=FR08-AI-049; title=Success response shape observation for a documented checkout request; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}; recording the response status, Content-Type, and full body; expected=The specification defines no success status or response schema, so this case asserts only that the response is JSON-parseable and records its exact fields and types as the observed contract; the officially expected status and schema remain SPEC GAP and no field may be asserted as required.	INVALID	The request is byte-identical to FR08-AI-001 and the case asserts no invariant; recording an undocumented success response shape is evidence attached to an existing case, not a separate product test case. Evidence basis: exploratory; requirement references: API-SPEC-4.3, FR-08; source anchors preserved in provenance/test-cases.json .	Drop FR08-AI-049. Record the observed success status, Content-Type, and body fields as part of the FR08-AI-001 execution evidence.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
opus-5) via Claude Code. Generated at: 2026-08-23T06:56:49.430Z.				
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B4, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: request envelope and body-shape faults, plus response-shape observation where the specification defines no exact schema. - Parameters/states/SEC/schema rules in scope: absent body, empty JSON object, malformed JSON, wrong top-level JSON type, structured wrong type for a documented field, unknown additional properties, request Content-Type handling, and observation of the actual success response shape." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:56:49.430Z.	Verbatim registry artifact excerpt: ID=FR08-AI-050; title=Rejection response shape observation for an unauthenticated checkout; request=POST /api/checkout with no Authorization header and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}, recording the response status, Content-Type, and full body; expected=SEC-02 requires refusal but no error schema is documented, so this case asserts only that the refusal is a structured JSON response and records its exact fields; the officially expected status and error schema remain SPEC GAP.	INVALID	The request is byte-identical to FR08-AI-019 and the state invariant is already asserted there, so the case only records an observation rather than testing behavior. Evidence basis: exploratory; requirement references: SEC-02, API-SPEC-4; source anchors preserved in provenance/test-cases.json .	Drop FR08-AI-050. Record the observed rejection status and error body as part of the FR08-AI-019 execution evidence.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B5, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: closure of the coverage ledger. Add only meaningful cases for risks the four earlier batches leave uncovered, and add nothing that merely restates an existing case. - Parameters/states/SEC/schema rules in scope: cart-quantity arithmetic boundaries, the client-supplied price field of the add-to-cart body as an input to the FR-08 recalculated total, duplicate submission of the same checkout, control characters inside ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:58:23.569Z.	Verbatim registry artifact excerpt: ID=FR08-AI-051; title=Cart-derived total for the minimum cart of one unit; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 100000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by GET /api/orders/:id for the created order; expected=Quantity 1 is the lower boundary of a purchasable cart, and FR-08 requires the total to be derived from the cart, so the persisted total must be exactly 100000; the exact response schema is SPEC GAP.	VALID	Executable as written from a stated starting state, one injected fault, and an oracle anchored to an authoritative FR-08, SEC, or API-specification rule with undocumented behavior kept as SPEC GAP; not a duplicate of another reviewed case. Evidence basis: official_contract; requirement references: FR-08, API-SPEC-4.2; source anchors preserved in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B5, held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: closure of the coverage ledger. Add	Verbatim registry artifact excerpt: ID=FR08-AI-052; title=Cart-derived total for a large quantity; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 9900000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by	INVALID	Quantity 99 is not a documented boundary, and replacing it with another arbitrary large quantity such as 99999 would still invent a boundary the authoritative specification does not define. Cart-derived arithmetic is already covered by the baseline, multi-line,	Drop FR08-AI-052 and do not create an arbitrary large-quantity replacement to preserve the case count. If later real runtime evidence reveals an actual numeric or quantity boundary, treat that separately.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
only meaningful cases for risks the four earlier batches leave uncovered, and add nothing that merely restates an existing case. - Parameters/states/SEC/schema rules in scope: cart-quantity arithmetic boundaries, the client-supplied <code>price</code> field of the add-to-cart body as an input to the FR-08 recalculated total, duplicate submission of the same checkout, control characters inside ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:58:23.569Z.	GET /api/orders/:id for the created order; expected=FR-08 requires the total to be derived from the cart, so the persisted total must be exactly 100000*99 = 9900000 with no overflow, truncation, or rounding; any documented stock or quantity ceiling is SPEC GAP.		aggregated-quantity, and minimum-cart cases. Evidence basis: official_contract_plus_exploration; requirement references: FR-08, API-SPEC-4.2; source anchors preserved in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B5 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: closure of the coverage ledger. Add only meaningful cases for risks the four earlier batches leave uncovered, and add nothing that merely restates an existing case. - Parameters/states/SEC/schema rules in scope: cart-quantity arithmetic boundaries, the client-supplied <code>price</code> field of the add-to-cart body as an input to the FR-08 recalculated total, duplicate submission of the same checkout, control characters inside ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:58:23.569Z.	Verbatim registry artifact excerpt: ID=FR08-AI-053; title=Client-supplied cart price as an input to the recalculated total; request=POST /api/cart {"id": 1, "name": "San pham A", "price": 1, "quantity": 1} with the user token, then POST /api/checkout with body {"total_amount": 1, "shipping_address": "123 Le Loi, TP.HCM"}, then GET /api/orders/:id for the created order; expected=FR-08 forbids the client from controlling the total; the add-to-cart body nevertheless carries a client-supplied price, so this case records whether the persisted total follows the catalogue price of product id 1 or the client-supplied 1, and whether the recalculation rule is defeated one level upst ... (truncated verbatim excerpt).	INCOMPLETE	The risk is useful, but the case has no source-grounded oracle about whether the cart must use the catalogue price. POST /api/cart exposes a client-supplied price, while FR-08 only states that checkout must derive the authoritative total from cart state. Evidence basis: official_contract_plus_exploration; requirement references: FR-08, API-SPEC-4.2, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json .	Keep the setup using a real product with a manipulated cart price. Before checkout, read back the actual server-observed cart state. FR-08 oracle: the persisted checkout total must equal the total derived from the cart state actually observed by the server immediately before checkout. Do not assert that checkout must replace the cart price with the catalogue price. Whether the cart endpoint should have accepted the manipulated price is an upstream specification-gap observation, not an FR-08 bug oracle.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B5 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: closure of the coverage ledger. Add only meaningful cases for risks the four earlier batches leave uncovered, and add nothing that merely restates an existing case. - Parameters/states/SEC/schema rules in scope: cart-quantity arithmetic boundaries, the client-supplied <code>price</code> field of the add-to-cart body as an input to the FR-08 recalculated total, duplicate submission of	Verbatim registry artifact excerpt: ID=FR08-AI-054; title=Duplicate submission of the same checkout request; request=POST /api/checkout twice with Authorization: Bearer <user JWT> and identical body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}, sent back to back with no read in between, followed by GET /api/orders/my-orders; expected=FR-08 clears the cart after a successful checkout, so one cart must not yield two paid orders; record whether the count becomes N+1 or N+2 and what total any second order carries; no idempotency or concurrency contract is	INVALID	Duplicate of FR08-AI-036: both execute the same checkout-after-checkout state transition and assert the same effective invariant, and because Newman runs sequentially, omitting the intermediate read changes nothing on the server. Evidence basis: official_contract_plus_exploration; requirement references: FR-08, API-SPEC-4.4; source anchors preserved in provenance/test-cases.json .	Drop FR08-AI-054 and keep FR08-AI-036. Do not relabel a sequential duplicate as a race-condition test; a genuine concurrency test would need parallel clients outside the agreed Postman/Newman toolchain.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
the same checkout, control characters inside ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:58:23.569Z.	documented, so the exact status of the second request is SPEC GAP.			
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B5 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: closure of the coverage ledger. Add only meaningful cases for risks the four earlier batches leave uncovered, and add nothing that merely restates an existing case. - Parameters/states/SEC/schema rules in scope: cart-quantity arithmetic boundaries, the client-supplied price field of the add-to-cart body as an input to the FR-08 recalculated total, duplicate submission of the same checkout, control characters inside ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:58:23.569Z.	Verbatim registry artifact excerpt: ID=FR08-AI-055; title=Control characters embedded in the shipping address; request=POST /api/checkout with Authorization: Bearer <user JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi\N\tTP.HCM"} where the address contains a carriage return, a line feed, and a tab as JSON escape sequences; expected=No character-set rule is documented for shipping_address; the case asserts only that the endpoint returns a structured HTTP response and that GET /api/orders/:id returns the address with the control characters preserved or consistently normalised rather than truncated at the first control character; ... (truncated verbatim excerpt).	INCOMPLETE	No authoritative character-set, normalization, or truncation rule exists for shipping_address, but the wording requires the value to be preserved or consistently normalised and not truncated, which is stronger than the specification. Evidence basis: exploratory; requirement references: FR-08, API-SPEC-4.3; source anchors preserved in provenance/test-cases.json .	Keep the exact value 123 Le Loi\N\tTP.HCM. Corrected oracle: observe acceptance or rejection; if accepted, record the exact read-back value; do not classify preservation, normalization, or truncation as a contract violation without an authoritative rule; if an order is created, the cart-derived-total invariant still applies. Character normalization behavior remains SPEC GAP.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B5 , held in full at provenance/README.md and in this candidate's generationPrompt field: "Required technique/category: closure of the coverage ledger. Add only meaningful cases for risks the four earlier batches leave uncovered, and add nothing that merely restates an existing case. - Parameters/states/SEC/schema rules in scope: cart-quantity arithmetic boundaries, the client-supplied price field of the add-to-cart body as an input to the FR-08 recalculated total, duplicate submission of the same checkout, control characters inside ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T06:58:23.569Z.	Verbatim registry artifact excerpt: ID=FR08-AI-056; title=Checkout performed with an admin-role token; request=POST /api/checkout with Authorization: Bearer <admin JWT> and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}, followed by GET /api/orders/my-orders with the admin token; expected=Checkout is not an admin-only route, and SEC-03 restricts only admin APIs, so an admin token is simply a valid authenticated identity here; the created order must be attributed to the admin account with the cart-derived total 200000 per FR-08; whether the product intends admins to purchase at all is ... (truncated verbatim excerpt).	INVALID	No authoritative source defines whether an admin user may purchase, so the case has no oracle beyond what FR08-AI-001 already proves. SEC-03 governs admin routes and the verified extract records it as not applicable to checkout, so the case would claim role-boundary coverage this API does not have. Evidence basis: exploratory; requirement references: FR-08, SEC-02, SEC-03; source anchors preserved in provenance/test-cases.json .	Drop FR08-AI-056. Do not invent an authorization rule for an undocumented admin-as-customer scenario, and do not claim SEC-03 coverage for checkout.
Verbatim excerpt of the stored bounded prompt for batch FR08-GEN-B6 , held in full at provenance/README.md and	Verbatim registry artifact excerpt: ID=FR08-AI-057; title=Validly signed but expired JWT must not authorize checkout;	VALID	Closes the documented expired-token authorization partition with a validly signed but expired fixture, so it fails for expiry rather than for	Accepted as-is. Preserve the expired token strictly as test input; no product oracle may be derived from source code.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
in this candidate's generationPrompt field: "Required technique/category: the single uncovered expired authorization partition. - Parameters/states/SEC/scheme rules in scope: SEC-02 valid-token requirement, the FR-08 logged-in rule, absence of order creation, and absence of cart mutation." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T07:16:05.889Z.	request=POST /api/checkout with header Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6MiwiYm9sZSI6InVzZXliLCJleHAiOiJF9.zG2FX6zN5zhGtWdl1e0AHVMg7iJ1IIIRT9vZbKCKIs and body {"total_amount": 200000, "shipping_address": "123 Le Loi, TP.HCM"}; expected=SEC-02 requires a valid JWT and FR-08 restricts checkout to logged-in users; an expired token is not valid, so checkout must not be authorized, the order count read back with the genuine token must remain N, and the cart must still contain product id 1 quantity 2; the exact rejection status and error ... (truncated verbatim excerpt).		signature validity and is therefore distinct from FR08-AI-022. Evidence basis: official_contract; requirement references: FR-08, SEC-02, API-SPEC-4; source anchors preserved in provenance/test-cases.json .	

3F. AI-Assisted FR-08 Postman implementation artifact

This row audits the executable FR-08 implementation produced after the student's canonical review and extension decisions. Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Date: 23/08/2026. The artifact was committed as `ec6db5d` before any official execution, so no oracle could be tuned to an observed result.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user instruction, verbatim excerpt from the conversation turn that authorised implementation: "Build the FR-08 executable suite from: the 51 reviewed-usable AI-origin cases; the five HUMAN extension cases. Ensure every canonical executable case is represented exactly once ... Set: X-Student-Id: 23127179 through the established collection-level mechanism so every request, including helpers, carries it. Do not invent expected HTTP status codes or response schemas. Use persisted-state/read-back assertions wherever the case has an authoritative invariant. For exploratory cases, capture observations without converting undocumented behavior into FAIL." Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Date: 23/08/2026. Exact chat time not separately captured; the committed artifact time is preserved in Git.	Verbatim artifact identifiers: collection= fr08/postman/FR08-checkout.postman_collection.json ; environment= fr08/postman/FR08-checkout.postman_environment.json ; data= fr08/postman/FR08-checkout.data.json with 56 rows; generator= fr08/postman/build-data.mjs ; commit= ec6db5d ; collection sha256= 7d185f739fa800f84d4433ddd7586928aaba225277e88b546984926e099de82 ; data sha256= 4b9e7c3824e7b78a1d15cda9271532019e062206a1bf9f6dd73a888ab6005b23 . Harness design recorded in the collection description: a fresh registered user per case, expected totals derived at run time from the cart the server itself reports immediately before each checkout, and no status or schema oracle anywhere.	VALID	The implementation satisfies every stated constraint and is verifiable against real evidence. Static preflight proved each reviewed-usable case is represented exactly once and no INVALID case is executable; the runner proved X-Student-Id: 23127179 on 438/438 executed requests; the triage in provenance/README.md found zero harness defects, zero state contamination, and zero blocked cases, so all 19 failures were product behaviour rather than implementation error. Evidence: provenance/run-index.md (RUN-20260823080014785-fr08-canonical-full-suite)/ and provenance/README.md .	The student required the implementation to be committed before execution, later required a Postman Desktop manual-import package under fixed display names, imported it by hand, ran the Collection Runner against the same data file, and attested the three resulting screenshots.

3G. AI-Assisted FR-08 GitHub Issue artifacts

These rows audit the two public bug-report artifacts drafted after the student confirmed both defects and then explicitly authorised publication. Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Date: 23/08/2026. The canonical publication timestamps are retained in `provenance/bugs.json`; chat interaction times were not separately captured.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user publication authorisation, verbatim: "Tôi cho phép publish cả hai Issue FR-08: BUG-CANDIDATE-FR08-CLIENT-TOTAL và BUG-CANDIDATE-FR08-CART-NOT-CLEARED." The preceding turn required the body to state the FR-08 requirement, the preconditions, why the finding is genuine, the impact without asserting a severity the assignment does not define, and the scope note that coincidental passes prove nothing. Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Date: 23/08/2026.	Verbatim final artifact identifiers: title= [HW06][FR-08] POST /api/checkout persists the client-supplied total_amount instead of recalculating it from the cart; GitHub Issue #15; URL= https://github.com/BuhDuy256/automation-testing-hw04/issues/15; canonical cases in the body= FR08-AI-002, FR08-AI-003, FR08-AI-004, FR08-AI-005, FR08-AI-035, FR08-H-001; evidence commit= 713bad65fb85966e9eb9febf2627a0d228bd8d11.	VALID	The published artifact is traceable to the explicit FR-08 rule that the backend recalculates the total and does not accept the client value, to the canonical run, and to attested evidence. It asserts no HTTP status, claims no severity level, and carries the recorded evidence limitation that coincidental passes do not prove server-side derivation. <code>publish-bug.mjs</code> re-read the created Issue through <code>gh issue view</code> and verified state, title, bug id, case ids, and rendered image markdown before persisting the number. Evidence: <code>provenance/bugs.json</code> , <code>provenance/evidence.json</code> , and <code>RUN-20260823080014785-fr08-canonical-full-suite</code> .	The student confirmed the defect at the bug gate, directed that all failing partitions stay grouped under one root-defect record instead of being split, inspected the published Issue page, and attested <code>EVID-FR08-CLIENT-TOTAL-ISSUE</code> .
Actual user publication authorisation, verbatim: "Tôi cho phép publish cả hai Issue FR-08: BUG-CANDIDATE-FR08-CLIENT-TOTAL và BUG-CANDIDATE-FR08-CART-NOT-CLEARED." The preceding turn additionally required secondary failures caused by the same defect to stay grouped rather than become separate bugs. Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Date: 23/08/2026.	Verbatim final artifact identifiers: title= [HW06][FR-08] The cart is not cleared after a successful checkout; GitHub Issue #16; URL= https://github.com/BuhDuy256/automation-testing-hw04/issues/16; canonical case in the body= FR08-AI-031; downstream contamination named for FR08-AI-040 and FR08-H-005; evidence commit= 713bad65fb85966e9eb9febf2627a0d228bd8d11.	VALID	The body stays inside the observed evidence: a 200 <code>Checkout successful</code> response with <code>orderId 27</code> , followed by a cart read-back still returning the purchased line, against the explicit FR-08 clearing rule. It presents the two downstream failures as second-order consequences rather than separate findings, and asserts no status code or severity. Verified after creation through <code>gh issue view</code> . Evidence: <code>provenance/bugs.json</code> and the canonical run.	The student confirmed the defect, required the consequence failures to remain grouped under this single record, inspected the published Issue page, and attested <code>EVID-FR08-CART-NOT-CLEARED-ISSUE</code> .

3H. AI-Assisted FR-08 CI workflow artifact

This row audits the FR-08 GitHub Actions demonstration. Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Date: 23/08/2026. Commit and GitHub Actions timestamps are canonical external provenance.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user continuation instruction, verbatim excerpt: "Proceed automatically through all actions that do not require genuine human judgment", together with the standing rules "Do not weaken an oracle to make the suite green" and "Never modify a product oracle merely because the SUT failed it". Tool: Claude Opus	Verbatim artifact/result identifiers: workflow= https://github.com/BuhDuy256/automation-testing-hw04/blob/hw06-api-testing/.github/workflows/hw06-fr08-ci.yml; demonstration case= FR08-AI-019; all-pass run= 32629097098 at commit a1f2188, conclusion success, 0 failed cases;	INCOMPLETE	The workflow and its two real runs validly demonstrate CI mechanics, artifact preservation, one genuinely green result and exactly one transparent deliberate failure, and the chosen case is one that truly passes rather than one that hides a published defect. The output is nevertheless incomplete against the assignment phrase "all API test cases	The student attested both CI screenshots. The decision not to build an FR-08 canonical full-suite workflow was declared openly as a reversible scope decision in <code>submission-checklist.md</code> rather than presented as full compliance, and the equivalent full-suite artifact remains the FR-04 one.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
5 (claude-opus-5) via Claude Code. Date: 23/08/2026.	intentional-failure run= 32629191161 at commit 87054fd , conclusion failure , exactly 1 failed case; intentional assertion= FR08-AI-019 [CI-DEMO] intentional single failure ; local pre-check= RUN-20260823084429748-fr08-ci-demo-local , exit code 0, 8/8 assertions.		passing": the sample executes only FR08-AI-019 , and no FR-08 canonical full-suite workflow was built, so CI does not prove that all 56 executable FR-08 cases pass. capture-ci-run.mjs refused to record any run whose remote conclusion and failure count contradicted the declared purpose. Evidence: provenance/ci-runs.json and the two attested run screenshots.	

3I. AI-Generated FR-15 Test-Case Artifacts

This section adds the 59 immutable AI candidates for FR-15 `POST /api/products` . Batches B1-B5 record Claude Opus 5 (`claude-opus-5`) via Claude Code; the closure batch B6 records OpenAI Codex (GPT-5). Every row uses the candidate's stored verbatim prompt, actual generation timestamp, original output fields, and the student's canonical review. The five `origin=HUMAN` extension cases are not counted as AI test artifacts.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-001; title=Create a product with all documented fields valid; request=POST /api/products with Authorization: Bearer <admin JWT>, Content-Type: application/json, body {"name": "FR15 AI 001 <run-unique suffix>", "price": 100000, "description": "FR15 baseline product", "imageUrl": "https://placeholder.co/300x300/png?text=FR15", "category_id": 1}; expected=FR-15 requires a non-empty name, a price greater than zero and an existing category, all of which hold, so the product is created and appears in GET /api/products with exactly the submitted name, price and category_id; the exact success status and response schema are SPEC GAP.	VALID	Executable as written with a distinct domain_partition and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15, FR-12, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's <pre>generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1

Batch identity

- Generation batch ID: FR15-GEN-B1
- Coverage stage: contract- domain
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude- opus-5) via Claude Code. Generated at: 2026-08- 23T08:55:15.640Z.</pre>	Verbatim registry artifact excerpt: ID=FR15-AI-002; title=Name at the documented maximum length of 255 characters; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "<the ASCII character N repeated exactly 255 times>", "price": 100000, "description": "FR15 name upper boundary", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=FR-15 allows a name of at most 255 characters, so 255 is the last accepted value and the product must be created with the full 255-character name preserved on read-back; the exact success status is SPEC GAP.	INCOMPLETE	The 255-character value is inside the documented limit, but exact full-length storage is not an authoritative FR-15 requirement. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Keep the 255-character input. Assert only creation because 255 is allowed; record the stored value and length. Do not classify truncation or normalization as a contract defect without an authoritative rule.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's <pre>generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1

Batch identity

- Generation batch ID: FR15-GEN-B1
- Coverage stage: contract- domain
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude- opus-5) via Claude Code. Generated at: 2026-08- 23T08:55:15.640Z.</pre>	Verbatim registry artifact excerpt: ID=FR15-AI-003; title=Name one character above the documented maximum; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "<the ASCII character N repeated exactly 256 times>", "price": 100000, "description": "FR15 name over boundary", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=FR-15 caps the name at 255 characters, so a 256-character name must not become a persisted product; assert through GET /api/products that no product carrying that name exists; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct domain_partition and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1, retained in full in this candidate's <pre>generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1

Batch identity

- Generation batch ID: FR15-GEN-B1
- Coverage stage: contract- domain
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude- opus-5) via Claude Code. Generated at: 2026-08- 23T08:55:15.640Z.</pre>	Verbatim registry artifact excerpt: ID=FR15-AI-004; title=Name of exactly one character; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "F", "price": 100000, "description": "FR15 name lower boundary", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=FR-15 requires only that the name is non-empty and at most 255 characters, so a single character is the smallest valid name and the product must be created with that exact name; the exact success status is SPEC GAP.	VALID	Executable as written with a distinct domain_partition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1, retained in full in this candidate's <pre>generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1

Batch identity

- Generation batch ID: FR15-GEN-B1
- Coverage stage: contract- domain
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude- opus-5) via Claude Code. Generated at: 2026-08- 23T08:55:15.640Z.</pre>	Verbatim registry artifact excerpt: ID=FR15-AI-005; title=Empty-string name; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "", "price": 100000, "description": "FR15 empty name", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=FR-15 makes the product name required, so an empty name must not become a persisted product; assert through GET /api/products that no product with an empty name was added; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct domain_partition and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1, retained in full in this candidate's <pre>generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1

Batch identity

- Generation batch ID: FR15-GEN-B1
- Coverage stage: contract-</pre>	Verbatim registry artifact excerpt: ID=FR15-AI-006; title=Whitespace-only name; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": " ", "price": 100000, "description": "FR15 whitespace name", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1} with exactly three space	VALID	Executable as written with a distinct domain_partition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: FR-15; source anchors are retained in	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	characters as the name; expected=Whether a whitespace-only name counts as empty under the documented required-name rule is not stated, so the case records whether such a product is persisted and what name is stored; the exact status and any trimming rule are SPEC GAP.		provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-007; title=Name field omitted from the request body; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"price": 100000, "description": "FR15 missing name", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1} with no name key; expected=FR-15 makes the name required, so an omitted name must not produce a persisted product; assert through GET /api/products that no nameless product was added; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct domain_partition and schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool:	Verbatim registry artifact excerpt: ID=FR15-AI-008; title=Name sent with a numeric type; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": 12345, "price": 100000, "description": "FR15 wrong-type name", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=The specification shows the name as text but defines no type-validation rule, so the case records whether a non-string name is rejected, coerced, or stored; the exact status is SPEC GAP.	VALID	Executable as written with a distinct domain_partition and schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.				
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-009; title=Price equal to zero; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 009 <run-unique suffix>", "price": 0, "description": "FR15 zero price", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=FR-15 requires a price greater than zero, so zero is the nearest invalid boundary and must not become a persisted product; assert through GET /api/products that no product with that unique name exists; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct domain_partition and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-010; title=Negative price; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 010 <run-unique suffix>", "price": -100000, "description": "FR15 negative price", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=A negative price violates the documented rule that the price must be greater than zero, so no product with that unique name may be persisted; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct domain_partition and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-	Verbatim registry artifact excerpt: ID=FR15-AI-011; title=Smallest positive price; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 011 <run-unique suffix>", "price": 1,	INCOMPLETE	Price 1 is a concrete positive integer, but FR-15 defines a positive number and documents no minimum increment or precision, so it is not the mathematical smallest-positive boundary. Evidence basis:	Keep price=1 as one concrete positive integer example. Remove "smallest positive" and boundary language; record the stored value without inventing precision or increment rules.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	"description": "FR15 minimum positive price", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=A price of 1 satisfies the documented greater-than-zero rule, so the product must be created and read back with the stored price exactly 1; the exact success status is SPEC GAP.		official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-012; title=Fractional positive price; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 012 <run-unique suffix>", "price": 0.5, "description": "FR15 fractional price", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=0.5 is greater than zero and therefore satisfies the documented rule; the case records whether the fractional value is stored exactly, rounded, or rejected, since no precision rule is documented; the exact status and any rounding behaviour are SPEC GAP.	VALID	Executable as written with a distinct domain_partition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice	Verbatim registry artifact excerpt: ID=FR15-AI-013; title=Price sent as a numeric string; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 013 <run-unique suffix>", "price": "100000", "description": "FR15 string price", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=The specification shows the price as a number and defines no coercion rule, so the case records whether a numerically parseable string is accepted and what value is stored; if a product is created its stored price must	VALID	Executable as written with a distinct domain_partition and schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	still be a positive value per FR-15; the exact status is SPEC GAP.			
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-014; title=Price sent as a non-numeric string; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 014 <run-unique suffix>", "price": "abc", "description": "FR15 non-numeric price", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=A non-numeric price cannot satisfy the documented greater-than-zero rule, so no product with that unique name may be persisted with such a price; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct domain_partition and schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-015; title=Price field omitted from the request body; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 015 <run-unique suffix>", "description": "FR15 missing price", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1} with no price key; expected=FR-15 requires a price greater than zero, so a request without any price must not create a persisted product; assert through GET /api/products that the unique name is absent; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct domain_partition and schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 ,	Verbatim registry artifact excerpt: ID=FR15-AI-016; title=Price explicitly null;	VALID	Executable as written with a distinct domain_partition and schema_validation purpose,	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 016 <run-unique suffix>", "price": null, "description": "FR15 null price", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=A null price cannot be greater than zero, so no product carrying that unique name and a null price may be persisted under FR-15; the exact rejection status and error schema are SPEC GAP.		concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-017; title=Nonexistent category id; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 017 <run-unique suffix>", "price": 100000, "description": "FR15 ghost category", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 9999}; expected=FR-15 requires the selected category to exist, so a product referencing a nonexistent category must not be persisted; assert through GET /api/products that the unique name is absent; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct domain_partition and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion	Verbatim registry artifact excerpt: ID=FR15-AI-018; title=Category id omitted from the request body; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 018 <run-unique suffix>", "price": 100000, "description": "FR15 missing category", "imageUrl": "https://placeholder.co/300x300/png"} with no category_id key; expected=FR-15 requires an existing category,	VALID	Executable as written with a distinct domain_partition and schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	so a request with no category at all must not create a persisted product; assert through GET /api/products that the unique name is absent; the exact rejection status and error schema are SPEC GAP.			
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-019; title=Category id sent as a non-numeric string; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 019 <run-unique suffix>", "price": 100000, "description": "FR15 wrong-type category", "imageUrl": "https://placeholder.co/300x300/png", "category_id": "laptop"}; expected=A non-numeric category reference cannot identify an existing category, so under FR-15 no product with that unique name may be persisted; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct domain_partition and schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code.	Verbatim registry artifact excerpt: ID=FR15-AI-020; title=Second valid seeded category; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 020 <run-unique suffix>", "price": 250000, "description": "FR15 second category", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 2}; expected=Category 2 is an existing category, so the create satisfies FR-15 and the product must be read back with category_id exactly 2, proving the category reference is stored as submitted rather than defaulted; the exact success status is SPEC GAP.	VALID	Executable as written with a distinct domain_partition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Generated at: 2026-08-23T08:55:15.640Z.				
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-021; title=Empty description; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 021 <run-unique suffix>", "price": 100000, "description": "", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=No validation rule is documented for the description, so the case records whether an empty description is accepted and what value is stored; the exact status is SPEC GAP.	VALID	Executable as written with a distinct domain_partition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-022; title=Description omitted from the request body; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 022 <run-unique suffix>", "price": 100000, "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1} with no description key; expected=The optionality of the description is not specified, so the case records whether the create succeeds without it and what description value is stored; the exact status is SPEC GAP.	VALID	Executable as written with a distinct domain_partition and schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-023; title=Description of 5000 characters; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 023 <run-unique suffix>", "price": 100000, "description": "<the ASCII	VALID	Executable as written with a distinct domain_partition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	character D repeated exactly 5000 times>", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=No maximum length is documented for the description, so the case observes acceptance or rejection and, if a product is created, records the stored description length; truncation is recorded rather than judged because no rule defines it; the exact status is SPEC GAP.		anchors are retained in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-024; title=Malformed imageUrl value; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 024 <run-unique suffix>", "price": 100000, "description": "FR15 malformed image url", "imageUrl": "not-a-valid-url", "category_id": 1}; expected=No URL-validation rule is documented for imageUrl, so the case records whether a malformed value is accepted and what is stored; the exact status and any URL rule are SPEC GAP.	VALID	Executable as written with a distinct domain_partition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed	Verbatim registry artifact excerpt: ID=FR15-AI-025; title=imageUrl omitted from the request body; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 025 <run-unique suffix>", "price": 100000, "description": "FR15 missing image url", "category_id": 1} with no imageUrl key; expected=The optionality of imageUrl is not specified, so the case records whether the create succeeds without it and what value is stored in its place; the exact status is SPEC GAP.	VALID	Executable as written with a distinct domain_partition and schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.				
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B1 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B1 ## Batch identity - Generation batch ID: FR15-GEN-B1 - Coverage stage: contract-domain - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Au ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:55:15.640Z.	Verbatim registry artifact excerpt: ID=FR15-AI-026; title=Vietnamese Unicode name round-trip; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "Điện thoại thử nghiệm FR15 <run-unique suffix>", "price": 100000, "description": "Mô tả sản phẩm kiểm thử", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1} written with real Vietnamese diacritics; expected=A Vietnamese name is inside the valid partition of the documented name rule, so the product must be created and the name read back character-identical with no encoding corruption; the exact success status is SPEC GAP.	INCOMPLETE	Vietnamese Unicode input is useful, but FR-15 does not require exact character-for-character round-trip fidelity or define normalization behavior. Evidence basis: official_contract_plus_exploration; requirement references: FR-15, SEC-04; source anchors are retained in provenance/test-cases.json .	Keep the Vietnamese input and record the stored/read-back value. Treat normalization and exact fidelity as SPEC GAP unless an authoritative rule is found.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2 ## Batch identity - Generation batch ID: FR15-GEN-B2 - Coverage stage: authorization-security - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:57:07.677Z.	Verbatim registry artifact excerpt: ID=FR15-AI-027; title=Create a product with no Authorization header; request=POST /api/products with no Authorization header and body {"name": "FR15 AI 027 <run-unique suffix>", "price": 100000, "description": "FR15 unauthenticated create", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=FR-12 restricts product data changes to admins and SEC-02 requires a valid JWT on secured APIs, so the request must be refused and GET /api/products must not contain the unique name; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct security and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-12, FR-15, SEC-02; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's	Verbatim registry artifact excerpt: ID=FR15-AI-028; title=Create a product with a valid ordinary-user token; request=POST /api/products	VALID	Executable as written with a distinct security and state_transition purpose, concrete input, and a source-grounded or explicitly	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2 ## Batch identity - Generation batch ID: FR15-GEN-B2 - Coverage stage: authorization-security - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:57:07.677Z.	with Authorization: Bearer <ordinary user JWT> and body {"name": "FR15 AI 028 <run-unique suffix>", "price": 100000, "description": "FR15 non-admin create", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=SEC-03 requires the admin API to check role = admin rather than merely the presence of a token, and FR-12 restricts product changes to admins, so a valid ordinary-user token must be refused and GET /api/products must not contain the unique name; the exact rejection status and error schema are SPEC GAP.		observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-12, FR-15, SEC-03; source anchors are retained in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2 ## Batch identity - Generation batch ID: FR15-GEN-B2 - Coverage stage: authorization-security - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:57:07.677Z.	Verbatim registry artifact excerpt: ID=FR15-AI-029; title=Create a product with a non-Bearer authorization scheme; request=POST /api/products with header Authorization: Token abc123 and body {"name": "FR15 AI 029 <run-unique suffix>", "price": 100000, "description": "FR15 wrong scheme", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=SEC-02 and the documented Bearer header rule require a valid JWT presented in the Bearer scheme, so a non-Bearer scheme must be refused and no product with the unique name may exist; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct security and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: SEC-02, FR-12; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2 ## Batch identity - Generation batch ID: FR15-GEN-B2 - Coverage stage: authorization-security - The invoking session records the actual tool/model and completion time immediately after the	Verbatim registry artifact excerpt: ID=FR15-AI-030; title=Create a product with a structurally invalid token; request=POST /api/products with header Authorization: Bearer not-a-jwt and body {"name": "FR15 AI 030 <run-unique suffix>", "price": 100000, "description": "FR15 invalid token", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=A string that is not a three-segment JWT cannot be a valid token, so SEC-02 requires refusal with a structured HTTP response	VALID	Executable as written with a distinct security and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: SEC-02, FR-12; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:57:07.677Z.	and no product with the unique name may exist; the exact rejection status and error schema are SPEC GAP.			
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2 ## Batch identity - Generation batch ID: FR15-GEN-B2 - Coverage stage: authorization-security - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:57:07.677Z.	Verbatim registry artifact excerpt: ID=FR15-AI-031; title=Create a product with an admin token whose signature was tampered with; request=POST /api/products with Authorization: Bearer <admin token with the deterministic final-character signature mutation> and body {"name": "FR15 AI 031 <run-unique suffix>", "price": 100000, "description": "FR15 tampered signature", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=SEC-02 requires token validity to be verified, so a token failing signature verification must be refused and no product with the unique name may exist; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct security and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: SEC-02, FR-12; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2 ## Batch identity - Generation batch ID: FR15-GEN-B2 - Coverage stage: authorization-security - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:57:07.677Z.	Verbatim registry artifact excerpt: ID=FR15-AI-032; title=Create a product with a validly signed but expired admin token; request=POST /api/products with Authorization: Bearer <validly signed but expired JWT fixture> and body {"name": "FR15 AI 032 <run-unique suffix>", "price": 100000, "description": "FR15 expired token", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=An expired token is not valid, so SEC-02 and FR-12 require refusal and GET /api/products must not contain the unique name; failing for expiry rather than for signature validity is what distinguishes this case from FR15-AI-031; the exact rejection status and error schema are SPEC GAP.	VALID	A validly signed but expired token is a distinct token-validity/security robustness partition under SEC-02, which requires a valid JWT; it remains distinct from signature tampering. Evidence basis: official_contract; requirement references: SEC-02, FR-12; source anchors are retained in provenance/test-cases.json .	Accepted with corrected grounding: cite SEC-02's valid-JWT requirement. Do not claim the verified extract explicitly lists expired JWT as an invalid partition.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2 ## Batch identity - Generation batch ID: FR15-GEN-B2 - Coverage stage: authorization-security - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:57:07.677Z.	Verbatim registry artifact excerpt: ID=FR15-AI-033; title=Client-supplied role field must not grant product creation; request=POST /api/products with Authorization: Bearer <ordinary user JWT> and body {"name": "FR15 AI 033 <run-unique suffix>", "price": 100000, "description": "FR15 client role escalation", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1, "role": "admin"}; expected=SEC-03 requires the role to be checked in the token, not taken from the request, so an extra body field claiming admin must not authorize the create and GET /api/products must not contain the unique name; the exact rejection status and any additional-property policy are SPEC GAP.	VALID	Executable as written with a distinct security and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: SEC-03, FR-12; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2 ## Batch identity - Generation batch ID: FR15-GEN-B2 - Coverage stage: authorization-security - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:57:07.677Z.	Verbatim registry artifact excerpt: ID=FR15-AI-034; title=HTML payload in the product name is stored and returned unchanged by the API; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": " <script>alert('xss') </script>FR15 AI 034 <run-unique suffix>", "price": 100000, "description": "FR15 script payload", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=SEC-04 places the escaping obligation at the UI display boundary, not at the API, so the API-side purpose is limited to preserving the value through the API: if the create is accepted the read-back value is character-identical, and if it is rejected no product is created. Whether the payload is rejected at the API is SPEC GAP, and this case neither proves nor disproves safe UI escaping.	INCOMPLETE	The HTML-like input is useful for observation, but API acceptance or read-back cannot prove SEC-04 display escaping, and FR-15 defines no character-preservation or normalization rule. Evidence basis: exploratory; requirement references: SEC-04, FR-15; source anchors are retained in provenance/test-cases.json .	Keep the HTML-like input. Observe acceptance/rejection and stored/read-back value without requiring character-identical preservation or treating transformation as an FR-15 defect. UI/display-boundary SEC-04 evidence remains separate.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2 ## Batch identity - Generation batch ID: FR15-GEN-B2 -	Verbatim registry artifact excerpt: ID=FR15-AI-035; title=SQL metacharacter payload in the product name must not damage product persistence; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 035 <run-unique suffix>"); DROP TABLE products; --", "price": 100000,	VALID	Executable as written with a distinct security purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: SEC-05, FR-15; source anchors are retained in	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Coverage stage: authorization-security - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:57:07.677Z.	"description": "FR15 sql payload", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=SEC-05 requires parameterized queries; the observable black-box oracle is that the endpoint returns a structured HTTP response and that GET /api/products still returns the previously seeded products afterwards, so no schema damage occurred. A safe response does not prove parameterized queries internally, and the exact status is SPEC GAP.		provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2 ## Batch identity - Generation batch ID: FR15-GEN-B2 - Coverage stage: authorization-security - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:57:07.677Z.	Verbatim registry artifact excerpt: ID=FR15-AI-036; title=SQL metacharacter payload in the category reference must not damage product persistence; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 036 <run-unique suffix>", "price": 100000, "description": "FR15 sql category payload", "imageUrl": "https://placeholder.co/300x300/png", "category_id": "1 OR 1=1"}; expected=Per FR-15 a product may only reference an existing category, so this value must not create a product with a valid category, and per SEC-05 the endpoint must return a structured response with GET /api/products still functioning and the seeded products intact; the exact status is SPEC GAP and the result cannot prove the internal query technique.	VALID	Executable as written with a distinct security and domain_partition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: SEC-05, FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2 ## Batch identity - Generation batch ID: FR15-GEN-B2 - Coverage stage: authorization-security - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs.	Verbatim registry artifact excerpt: ID=FR15-AI-037; title=Refused non-admin create must not alter the existing catalogue; request=POST /api/products with Authorization: Bearer <ordinary user JWT> and body {"name": "FR15 AI 037 <run-unique suffix>", "price": 100000, "description": "FR15 catalogue integrity", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=A refused create is not a data change, so beyond creating no new product it must leave every previously existing product unchanged in id, name and price when GET /api/products is read back; the exact rejection status is SPEC GAP.	VALID	Executable as written with a distinct security and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: FR-12, SEC-03; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:57:07.677Z.				
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B2 , retained in full in this candidate's <pre>generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B2

Batch identity

- Generation batch ID: FR15-GEN-B2
- Coverage stage: authorization- security
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026- 08-23T08:57:07.677Z.</pre>	Verbatim registry artifact excerpt: ID=FR15-AI-038; title=Admin token remains sufficient for repeated creates within one session; request=POST /api/products twice with the same Authorization: Bearer <admin JWT>, first with body {"name": "FR15 AI 038 A <run-unique suffix>", "price": 100000, "description": "FR15 repeat one", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1} and then with body {"name": "FR15 AI 038 B <run-unique suffix>", "price": 200000, "description": "FR15 repeat two", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=FR-12 authorizes admins to change product data, and nothing documents a single-use restriction, so both creates must succeed under the same token and both products must appear in GET /api/products with their own submitted names and prices; the exact success status is SPEC GAP.	INVALID	Repeats baseline creation and invents a token-reuse risk; multi-product persistence is already covered by FR15-AI-040 and FR15-AI-041. Evidence basis: official_contract_plus_exploration; requirement references: FR-12, FR-15; source anchors are retained in provenance/test-cases.json .	Drop FR15-AI-038; do not retain a repeated baseline create for count optics.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B3 , retained in full in this candidate's <pre>generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B3

Batch identity

- Generation batch ID: FR15-GEN-B3
- Coverage stage: state- transition
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## A ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026- 08-23T08:59:51.734Z.</pre>	Verbatim registry artifact excerpt: ID=FR15-AI-039; title=Created product is retrievable by its own detail endpoint; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 039 <run-unique suffix>", "price": 150000, "description": "FR15 detail read-back", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}, followed by GET /api/products/:id using the returned identifier; expected=A successful create is a persistence transition, so the detail endpoint must return the same product with exactly the submitted name, price 150000 and category_id 1; the exact detail response schema is SPEC GAP, and if the create returns no identifier the traceability failure is recorded as an evidence gap.	VALID	Executable as written with a distinct state_transition and schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: FR-15, API-SPEC-3.2, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B3 , retained in full in this	Verbatim registry artifact excerpt: ID=FR15-AI-040; title=Catalogue grows by exactly one product per	VALID	Executable as written with a distinct state_transition purpose, concrete input, and a source-grounded or	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B3

Batch identity

- Generation batch ID: FR15-GEN-B3
- Coverage stage: state-transition
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## A ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.734Z.	successful create; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 040 <run-unique suffix>", "price": 100000, "description": "FR15 count transition", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}, followed by GET /api/products; expected=One create adds one product, so the count must become exactly N+1 and the added entry must be the one carrying the unique name; the exact response schema is SPEC GAP.		explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: FR-15, API-SPEC-3.1; source anchors are retained in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B3 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B3

Batch identity

- Generation batch ID: FR15-GEN-B3
- Coverage stage: state-transition
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## A ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.734Z.	Verbatim registry artifact excerpt: ID=FR15-AI-041; title=A successful create leaves existing products unchanged; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 041 <run-unique suffix>", "price": 100000, "description": "FR15 non-interference", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}, followed by GET /api/products; expected=Creating one product must not modify any other product, so every previously captured product must still be present with identical id, name, price and category_id; the exact response schema is SPEC GAP.	VALID	Executable as written with a distinct state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: FR-15, API-SPEC-3.1; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B3 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B3

Batch identity

- Generation batch ID: FR15-GEN-B3
- Coverage stage: state-transition
- The invoking session records the actual tool/model and completion time immediately	Verbatim registry artifact excerpt: ID=FR15-AI-042; title=Created product is findable through the documented search query; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15AI042<run-unique suffix>", "price": 100000, "description": "FR15 search read-back", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}, followed by GET /api/products?	VALID	Executable as written with a distinct state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: FR-15, API-SPEC-3.1; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## A ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.734Z.	search=FR15AI042<run-unique suffix>; expected=The product list documents a search query by product name, so the newly created product must be returned by a search for its own unique name; the exact search response schema is SPEC GAP.			
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B3 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B3 ## Batch identity - Generation batch ID: FR15-GEN-B3 - Coverage stage: state-transition - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## A ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.734Z.	Verbatim registry artifact excerpt: ID=FR15-AI-043; title=Two products created with an identical name; request=POST /api/products twice with Authorization: Bearer <admin JWT> and the identical body {"name": "FR15 AI 043 <run-unique suffix>", "price": 100000, "description": "FR15 duplicate name", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}, followed by GET /api/products?search=; expected=No uniqueness rule is documented for product names, so the case records whether the second create is accepted and how many products carry that name; nothing is asserted about rejection because no source defines it, and the exact status is SPEC GAP.	VALID	Executable as written with a distinct state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B3 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B3 ## Batch identity - Generation batch ID: FR15-GEN-B3 - Coverage stage: state-transition - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## A ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.734Z.	Verbatim registry artifact excerpt: ID=FR15-AI-044; title=A valid create succeeds after a rejected invalid create with the same name; request=POST /api/products with Authorization: Bearer <admin JWT> and the invalid body {"name": "FR15 AI 044 <run-unique suffix>", "price": -1, "description": "FR15 recovery invalid", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}, then the valid body {"name": "FR15 AI 044 <run-unique suffix>", "price": 100000, "description": "FR15 recovery valid", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}, followed by GET /api/products; expected=The negative price violates the documented positive-price rule so the first attempt must not persist a product, and the following valid create must succeed, leaving exactly one product with that unique	VALID	Executable as written with a distinct state_transition and domain_partition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
	name and a stored price of 100000; the exact rejection status is SPEC GAP.			
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B3 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B3 ## Batch identity - Generation batch ID: FR15-GEN-B3 - Coverage stage: state-transition - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## A ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.734Z.	Verbatim registry artifact excerpt: ID=FR15-AI-045; title=Category association is stored for a third distinct category; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 045 <run-unique suffix>", "price": 100000, "description": "FR15 third category", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 3}, followed by GET /api/products; expected=FR-15 requires the selected category to exist and category 3 does, so the product must be created and read back with category_id exactly 3; the exact response schema is SPEC GAP.	INVALID	Using category 3 repeats FR15-AI-020's existing-category/non-defaulting partition and adds no distinct diagnostic purpose. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Drop FR15-AI-045 and keep FR15-AI-020.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B3 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B3 ## Batch identity - Generation batch ID: FR15-GEN-B3 - Coverage stage: state-transition - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## A ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.734Z.	Verbatim registry artifact excerpt: ID=FR15-AI-046; title=Creating a product does not modify the category list; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 046 <run-unique suffix>", "price": 100000, "description": "FR15 category integrity", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 2}, followed by GET /api/categories; expected=Creating a product references a category but must not create, rename or remove one, so the category list read back must be identical to the captured snapshot; the exact response schema is SPEC GAP.	INVALID	Product creation not modifying the category list is not an authoritative FR-15 requirement, and the inferred side-effect check is too weak for a separate product case. Evidence basis: official_contract_plus_exploration; requirement references: FR-15, API-SPEC-3.4; source anchors are retained in provenance/test-cases.json .	Drop FR15-AI-046.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B4 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-	Verbatim registry artifact excerpt: ID=FR15-AI-047; title=Create request sent with no body at all; request=POST /api/products with Authorization: Bearer <admin JWT>, header Content-Type: application/json, and no	VALID	Executable as written with a distinct schema_validation and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis:	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
B4

Batch identity

- Generation batch ID: FR15-GEN-B4
- Coverage stage: schema
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## Authoritati ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.828Z.	request body bytes; expected=FR-15 requires a name, a positive price and an existing category, none of which can be present without a body, so no product may be persisted; assert through GET /api/products that the count is unchanged; the exact rejection status and error schema are SPEC GAP.		official_contract; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B4 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B4

Batch identity

- Generation batch ID: FR15-GEN-B4
- Coverage stage: schema
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## Authoritati ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.828Z.	Verbatim registry artifact excerpt: ID=FR15-AI-048; title=Create request with an empty JSON object body; request=POST /api/products with Authorization: Bearer <admin JWT>, Content-Type: application/json, and body {}; expected=All three required FR-15 fields are absent at once, so no product may be persisted; assert through GET /api/products that the count is unchanged and no nameless product appeared; the exact rejection status and error schema are SPEC GAP.	VALID	Executable as written with a distinct schema_validation and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B4 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B4

Batch identity

- Generation batch ID: FR15-GEN-B4
- Coverage stage: schema
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim	Verbatim registry artifact excerpt: ID=FR15-AI-049; title=Create request with syntactically malformed JSON; request=POST /api/products with Authorization: Bearer <admin JWT>, Content-Type: application/json, and the exact raw body {"name": "FR15 AI 049", "price": with the value, closing quote and brace omitted so the payload is not valid JSON; expected=Parser-failure behaviour is not documented, so the case asserts only that the endpoint returns a structured HTTP response rather than failing without one, and that no product is persisted from	INCOMPLETE	Malformed JSON is a distinct request-envelope fault, but the original narrative does not fix the exact bytes testers must send. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Send exactly the raw body {"name": "FR15 AI 049", "price": with no value or closing brace, and prevent client re-serialization. Assert no persistence; status and error schema remain SPEC GAP.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
execution,
observed responses, or product bugs.

## Authoritati ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.828Z.	an unparseable payload; the exact status and error schema are SPEC GAP.			
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B4 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B4

Batch identity

- Generation batch ID: FR15-GEN-B4
- Coverage stage: schema
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## Authoritati ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.828Z.	Verbatim registry artifact excerpt: ID=FR15-AI-050; title=Create request whose top-level JSON value is an array; request=POST /api/products with Authorization: Bearer <admin JWT>, Content-Type: application/json, and body [{"name": "FR15 AI 050 <run-unique suffix>", "price": 100000, "description": "FR15 array body", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}]; expected=The documented request body is a JSON object; a top-level array is a structural mismatch with no documented handling, so the case asserts a structured HTTP response and records whether any product with the unique name was persisted; the exact status is SPEC GAP.	VALID	Executable as written with a distinct schema_validation and state_transition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B4 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B4

Batch identity

- Generation batch ID: FR15-GEN-B4
- Coverage stage: schema
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## Authoritati ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.828Z.	Verbatim registry artifact excerpt: ID=FR15-AI-051; title=Description sent as a nested JSON object; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 051 <run-unique suffix>", "price": 100000, "description": {"short": "FR15", "long": "FR15 structured description"}, "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=The specification shows the description as text and defines no structured-object handling, so the case records whether the object is rejected, stringified, or stored; the exact status is SPEC GAP.	VALID	Executable as written with a distinct schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B4 , retained in full in this	Verbatim registry artifact excerpt: ID=FR15-AI-052; title=Unknown additional property in the create body;	VALID	Executable as written with a distinct schema_validation purpose, concrete input, and a source-grounded or	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B4

Batch identity

- Generation batch ID: FR15-GEN-B4
- Coverage stage: schema
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## Authoritati ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.828Z.	request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 052 <run-unique suffix>", "price": 100000, "description": "FR15 unknown property", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1, "stock_quantity": 99}; expected=No additional-property rule is documented, so the case records whether the unknown field is ignored or rejected and asserts only that it does not change the stored name, price or category of the created product; the exact status and any additional-property policy are SPEC GAP.		explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B4 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B4

Batch identity

- Generation batch ID: FR15-GEN-B4
- Coverage stage: schema
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs.

## Authoritati ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.828Z.	Verbatim registry artifact excerpt: ID=FR15-AI-053; title=Create body sent with a text/plain Content-Type; request=POST /api/products with Authorization: Bearer <admin JWT>, header Content-Type: text/plain, and the raw body {"name": "FR15 AI 053 <run-unique suffix>", "price": 100000, "description": "FR15 text plain", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=No Content-Type requirement is documented, so the case records whether the payload is parsed or ignored; if a product is created it must carry exactly the submitted name, price and category per FR-15 rather than empty or default values; the exact status and content-type policy are SPEC GAP.	VALID	Executable as written with a distinct schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B4 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B4

Batch identity

- Generation batch ID: FR15-GEN-B4
- Coverage stage: schema
- The invoking session records the actual tool/model and completion time immediately after the	Verbatim registry artifact excerpt: ID=FR15-AI-054; title=imageUrl sent as a boolean; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 054 <run-unique suffix>", "price": 100000, "description": "FR15 boolean image url", "imageUrl": true, "category_id": 1}; expected=The specification shows imageUrl as text and defines no type validation, so the case records whether a boolean is rejected, coerced,	VALID	Executable as written with a distinct schema_validation purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ## Authoritati ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T08:59:51.828Z.	or stored; if a product is created its name, price and category must still match the submitted values per FR-15; the exact status is SPEC GAP.			
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B5 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B5 ## Batch identity - Generation batch ID: FR15-GEN-B5 - Coverage stage: closure-deduplication - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T09:01:07.762Z.	Verbatim registry artifact excerpt: ID=FR15-AI-055; title=Category id zero; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 055 <run-unique suffix>", "price": 100000, "description": "FR15 zero category", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 0}; expected=FR-15 requires an existing category and 0 is not one, so no product carrying that unique name may be persisted; the exact rejection status and error schema are SPEC GAP.	INVALID	Category id 0 is another nonexistent-category value and duplicates FR15-AI-017; no authoritative minimum category-id boundary is documented. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Drop FR15-AI-055 and retain FR15-AI-017 for the nonexistent-category partition.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B5 , retained in full in this candidate's generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B5 ## Batch identity - Generation batch ID: FR15-GEN-B5 - Coverage stage: closure-deduplication - The invoking session records the actual tool/model and completion time immediately after the response. ## Task boundary Generate candidate API test cases only for the coverage slice stated below. Do not claim execution, observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude-opus-5) via Claude Code. Generated at: 2026-08-23T09:01:07.762Z.	Verbatim registry artifact excerpt: ID=FR15-AI-056; title=Extremely large price; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 056 <run-unique suffix>", "price": 9999999999999999, "description": "FR15 large price", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1}; expected=The value is greater than zero and therefore satisfies the documented rule, and no upper limit is documented, so if a product is created its stored price must be exactly 9999999999999999 with no overflow or truncation; the exact status and any numeric ceiling are SPEC GAP.	INVALID	The extremely large positive value is not a documented boundary; reducing its oracle to price greater than zero duplicates existing valid-price coverage. Evidence basis: official_contract_plus_exploration; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Drop FR15-AI-056 rather than inventing a numeric precision or magnitude boundary.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B5 , retained in full in this candidate's <pre>generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B5

Batch identity

- Generation batch ID: FR15-GEN-B5
- Coverage stage: closure- deduplication
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude- opus-5) via Claude Code. Generated at: 2026-08- 23T09:01:07.762Z.</pre>	Verbatim registry artifact excerpt: ID=FR15-AI-057; title=Name with leading and trailing whitespace; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": " FR15 AI 057 <run-unique suffix> ", "price": 100000, "description": "FR15 padded name", "imageUrl": "https://placeholder.co/300x300/png", "category_id": 1} with exactly three leading and three trailing spaces; expected=The padded name is non-empty and within 255 characters so it satisfies the documented rule; no trimming rule is documented, so the case records whether the stored name keeps or loses the padding rather than judging it; the exact status and any trimming behaviour are SPEC GAP.	VALID	Executable as written with a distinct domain_partition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: official_contract_plus_exploration; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B5 , retained in full in this candidate's <pre>generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B5

Batch identity

- Generation batch ID: FR15-GEN-B5
- Coverage stage: closure- deduplication
- The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate candidate API test cases only for the coverage slice stated below. Do not claim execution,
observed responses, or product bugs. ... (truncated verbatim excerpt)" Tool: Claude Opus 5 (claude- opus-5) via Claude Code. Generated at: 2026-08- 23T09:01:07.762Z.</pre>	Verbatim registry artifact excerpt: ID=FR15-AI-058; title=imageUrl of 2000 characters; request=POST /api/products with Authorization: Bearer <admin JWT> and body {"name": "FR15 AI 058 <run-unique suffix> ", "price": 100000, "description": "FR15 long image url", "imageUrl": "https://placeholder.co/300x300/png?text=<the ASCII character U repeated until the whole value is exactly 2000 characters>", "category_id": 1}; expected=No maximum length is documented for imageUrl, so the case observes acceptance or rejection and, if a product is created, records the stored value length; truncation is recorded rather than judged; the exact status and any length limit are SPEC GAP.	VALID	Executable as written with a distinct domain_partition purpose, concrete input, and a source-grounded or explicitly observational oracle; not duplicated by another retained FR-15 case. Evidence basis: exploratory; requirement references: FR-15, API-SPEC-3.3; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.
Verbatim excerpt of the stored bounded prompt for batch FR15-GEN-B6 , retained in full in this candidate's <pre>generationPrompt field: "# HW06 AI Generation Prompt — HW06-C-FR15-POST-PRODUCTS / FR15-GEN-B6

Batch identity

- Generation batch ID: FR15-GEN-B6
- Coverage stage: category-</pre>	Verbatim registry artifact excerpt: ID=FR15-AI-059; title=Category id explicitly set to null; request=POST /api/products with Authorization: Bearer <admin JWT>, Content-Type: application/json, and body {"name": "FR15 AI 059 <run-unique suffix> ", "price": 100000, "description": "FR15 null category", "imageUrl": "https://placeholder.co/300x300/png"}.	VALID	Explicit null closes the authoritative invalid category partition and is distinct from omitted, wrong-type, and concrete nonexistent-category inputs. Evidence basis: official_contract; requirement references: FR-15; source anchors are retained in provenance/test-cases.json .	Accepted as-is; no correction required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
null-closure
 The invoking session records the actual tool/model and completion time immediately after the response.

Task boundary

Generate exactly one candidate API test case for the coverage slice below. Do not modify or duplicate
 FR15-AI-001 through FR15-AI-058 ... (truncated verbatim excerpt)" Tool: OpenAI Codex (GPT-5). Generated at: 2026-08-23T09:53:46.147Z.	0/png", "category_id": null); expected=FR-15 requires an existing category, and null identifies no existing category, so no product carrying the unique name may be persisted; assert through GET /api/products that the name is absent; the exact rejection status and error schema are SPEC GAP.			

3J. AI-Assisted FR-15 Independent Review and Correction Artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user review-direction prompt, verbatim excerpt: "Continue only with HG-FR15-REV-01. Do not record human-reviews.json and do not start ACT-EXT-01 yet. I am not approving the current 52 VALID / 4 INCOMPLETE / 2 INVALID proposal as-is." The turn supplied exact corrections for FR15-AI-002, 011, 026, 046, 049, 055 and 056, retained 038/045 as INVALID, and required a fresh null-category case plus independent/adversarial review. Tool: OpenAI Codex; exact model and chat time were not separately persisted. Date: 23/08/2026.	Verbatim artifact/result identifiers: generated FR15-AI-059 in prospective batch FR15-GEN-B6 ; regenerated provenance/README.md ; proposed 50 VALID, 4 INCOMPLETE and 5 INVALID after source, duplicate and mechanical review; preserved FR15-AI-001..058 unchanged.	INCOMPLETE	The revised proposal correctly applied the supplied corrections and closed the authoritative category_id=null partition, but the student found two remaining review defects: FR15-AI-034 required observational API preservation rather than character-identical preservation, and FR15-AI-032 incorrectly claimed the verified extract explicitly listed expired JWT.	The student changed FR15-AI-034 from VALID to INCOMPLETE, corrected FR15-AI-032 source grounding while retaining VALID, and approved the final 49 VALID / 5 INCOMPLETE / 5 INVALID accounting. Those exact decisions were recorded separately in human-reviews.json .

3K. AI-Assisted FR-15 Human-Extension Artifacts

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
The exact standalone prompt that produced the shortlist was not separately captured in a canonical file. Preserved artifact-level context: after ACT-REV-01, propose distinct FR-15 HUMAN extension directions for HG-FR15-EXT-01 without canonicalizing them. Tool: OpenAI Codex; exact model/time unavailable.	Verbatim artifact identifiers: provenance/README.md proposed directions A-F, each with risk/gap, input/state, observable oracle, source, missed-coverage explanation and duplicate analysis.	VALID	The shortlist gave the student genuine alternatives without adding HUMAN cases prematurely. The later decision not to select F does not invalidate the proposal artifact.	The student explicitly selected A-E and rejected F. No sixth HUMAN case was created.
Actual user selection prompt, verbatim excerpt: "Continue from HG-FR15-EXT-01 . I reviewed provenance/README.md . I approve the following five extension proposals without modification: A B C D E. Do not select proposal F." Tool: OpenAI Codex; exact model/chat time were not	Verbatim artifact identifiers: FR15-H-001 through FR15-H-005 , each with origin=HUMAN and a humanExtensionRationale ; duplicate/adversarial comparison against all 54 usable AI-origin cases found no effective duplicate; final executable design count=59.	VALID	The materialization preserved the student's exact selection and IDs, kept the cases diagnostically distinct, and did not modify any original AI candidate.	The student supplied and approved the five directions, their required oracles, and the instruction that they be recorded as HUMAN origin. They are excluded from the AI-candidate accuracy counts.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
separately persisted. Date: 23/08/2026.				

3L. AI-Assisted FR-15 Postman Implementation Artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual implementation instruction, verbatim excerpt from the same extension-selection turn: "Then continue according to: provenance/README.md, submission-checklist.md, provenance/README.md, provenance/README.md. Proceed automatically into FR-15 Postman implementation and the following non-human-gated actions. Do not weaken any oracle to make execution pass. Do not convert SPEC GAP observations into failures." Tool: OpenAI Codex; exact model/chat time were not separately persisted. Date: 23/08/2026.	Verbatim artifact identifiers: collection= fr15/postman/FR15-products.postman_collection.json ; environment= fr15/postman/FR15-products.postman_environment.json ; data= fr15/postman/FR15-products.postman_data.json with 59 rows; builder= fr15/postman/build.mjs ; pre-execution commit= 316c847 ; collection sha256= 32216e5f0740e8b4bd2dee1156f879274e562d81e080f232a182f3117671ca6c .	VALID	Static checks proved all 59 reviewed executable cases were represented, all five INVALID cases were excluded, every helper request received the student header, and authoritative state oracles remained separate from observational SPEC GAP behavior. The implementation was committed before execution.	The student fixed the reviewed suite and extension directions before implementation and explicitly prohibited oracle weakening, fabricated HTTP requirements, and automatic human evidence attestation.

3M. AI-Assisted FR-15 Execution and Triage Artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual execution scope from the extension-selection turn, verbatim excerpt: "Proceed automatically into FR-15 Postman implementation and the following non-human-gated actions. Do not weaken any oracle to make execution pass. Do not convert SPEC GAP observations into failures." Tool: OpenAI Codex; exact model/chat time were not separately persisted. Date: 23/08/2026.	Verbatim evidence/result identifiers: canonical= RUN-20260823102155874-fr15-canonical-full-suite , 59 mapped cases, 16 PASS, 28 FAIL, 15 SPEC-GAP, 0 HARNESS, 0 BLOCKED; authorization reproduction= RUN-20260823102829114-fr15-auth-reproduction , 5/5 failed persistence oracles and 40/40 student-header requests; validation reproduction= RUN-20260823102841701-fr15-validation-reproduction , 6/6 failed persistence oracles and 52/52 student-header requests.	VALID	The output preserved immutable raw reports and exit code 1, separated assertion count from case classification, mapped every authoritative failure, retained observational cases as SPEC-GAP, and found no harness, setup, cleanup or blocked result. No canonical rerun or oracle tuning occurred after the failures were observed.	The student later reviewed the proposal grounded in these runs and explicitly confirmed the two root-defect groups without splitting their input partitions.

3N. AI-Assisted FR-15 Bug Analysis and Evidence Artifacts

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Preserved checkpoint instruction: continue automatically after execution until the next genuine HUMAN_GATE, without publishing GitHub Issues, creating human attestation, or updating the AI Audit Report. Tool: OpenAI Codex; exact standalone	Verbatim artifact identifiers: BUG-CANDIDATE-FR15-AUTHORIZATION ; BUG-CANDIDATE-FR15-VALIDATION ; non-canonical review packet= provenance/README.md ; both records remained candidate pending student judgment.	VALID	The proposal grouped failures by two coherent root defects, grounded expected behavior in FR-12/FR-15/SEC-02/SEC-03, used persistent state rather than invented response codes, and stopped for human confirmation before external publication.	The student reviewed the packet and explicitly confirmed both candidates as genuine bugs while requiring each root grouping to remain intact.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
prompt/model/time were not separately persisted.				
Actual user confirmation/evidence/audit prompt, verbatim excerpt: "Continue from HG-FR15-BUG-02 . I reviewed provenance/README.md . I explicitly confirm both FR-15 bug candidates as genuine bugs" and "Do not publish either GitHub Issue yet." The same turn explicitly authorized updating ai-audit-report.md and required an evidence visual-attestation gate. Tool: OpenAI Codex; exact model/chat time were not separately persisted. Date: 23/08/2026.	Verbatim artifact/result identifiers at this checkpoint: both bug records changed to human-confirmed at 2026-08-23T10:40:24.2861323Z ; all 28 canonical FAIL cases map to one of the two confirmed records; evidence ids= EVID-FR15-RUN-SUMMARY , EVID-FR15-AUTHORIZATION , EVID-FR15-VALIDATION ; representative cases= FR15-AI-028 and FR15-AI-009 ; all three PNGs remain humanAttestation=false .	VALID	The workflow records the student's actual decision, preserves the canonical and targeted runs, creates only the requested minimal evidence set from unmodified real Newman HTML, verifies PNG integrity and provenance, and stops before attestation or Issue publication.	The student supplied the genuine-bug decisions and authorized this audit update, but has not yet visually attested any FR-15 screenshot or authorized GitHub Issue publication.

30. AI-Assisted FR-15 Published GitHub Issue Artifacts

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user publication prompt: "I authorize publishing both FR-15 GitHub Issues: BUG-CANDIDATE-FR15-AUTHORIZATION and BUG-CANDIDATE-FR15-VALIDATION." Tool: OpenAI Codex; exact model/chat time were not separately persisted. Date: 23/08/2026.	Verbatim published artifact identifiers: BUG-CANDIDATE-FR15-AUTHORIZATION ; GitHub Issue #17 ; URL= https://github.com/BuhDuy256/automation-testing-hw04/issues/17 ; publication timestamp= 2026-08-23T10:59:45.638Z .	VALID	The published Issue preserves the human-confirmed authorization root defect, all ten mapped canonical failures, persistent-state oracle, real reproduction evidence, and the documented boundary that exact rejection status is unspecified. The live Issue was read back before its number and URL were stored.	The student explicitly authorized external publication and later visually attested EVID-FR15-AUTHORIZATION-ISSUE .
Actual user publication prompt: "I authorize publishing both FR-15 GitHub Issues: BUG-CANDIDATE-FR15-AUTHORIZATION and BUG-CANDIDATE-FR15-VALIDATION." Tool: OpenAI Codex; exact model/chat time were not separately persisted. Date: 23/08/2026.	Verbatim published artifact identifiers: BUG-CANDIDATE-FR15-VALIDATION ; GitHub Issue #18 ; URL= https://github.com/BuhDuy256/automation-testing-hw04/issues/18 ; publication timestamp= 2026-08-23T10:59:56.376Z .	VALID	The published Issue preserves the human-confirmed validation root defect, all eighteen mapped canonical failures, persistent-state oracle, real reproduction evidence, and the documented boundary that exact rejection status and error schema are unspecified. The live Issue was read back before its number and URL were stored.	The student explicitly authorized external publication and later visually attested EVID-FR15-VALIDATION-ISSUE .

3P. AI-Assisted FR-15 CI Demonstration Artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user CI prompt: "BUILD FR-15 CI — use FR15-AI-001 and intentionally fail only the labelled CI-demo assertion." Tool: OpenAI Codex; exact model/chat time were not separately persisted. Date: 23/08/2026.	Verbatim artifact/result identifiers: workflow= https://github.com/BuhDuy256/automation-testing-hw04/blob/hw06-api-testing/.github/workflows/hw06-fr15-ci.yml ; case= FR15-AI-001 ; all-pass run= 32636054520 with 0 failures; intentional	INCOMPLETE	The artifact fulfills the requested bounded CI demonstration and preserves the real product and cleanup oracles, but one selected passing case is not evidence that the full 59-case canonical FR-15 suite is green. The canonical suite intentionally retains 28	The student chose the exact case and labelled failure mechanism, inspected both run screenshots, and explicitly attested them. The report retains the limitation instead of presenting the sample as full-suite success.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
	run= 32636132200 with exactly one [CI-DEMO] failure; restored-green run= 32636274089 ; screenshot ids= EVID-CI-FR15-ALL-PASS and EVID-CI-FR15-SINGLE-FAILURE .		confirmed product-defect failures.	

3Q. AI-Assisted FR-15 Postman Evidence and Final Curation Artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user evidence prompt, verbatim excerpts: "Hình như ta đang thiếu Postman Evidence đúng ko?" and "Tôi có screenshots các images như lúc này tôi làm với FR08. Hãy review và update tên." Tool: OpenAI Codex; exact model/chat time were not separately persisted. Date: 23/08/2026.	Verbatim artifact/result identifiers: EVID-FR15-POSTMAN-COLLECTION , EVID-FR15-POSTMAN-CONSOLE , and EVID-FR15-POSTMAN-RUNNER ; provenance/README.md ; curated fr15/ containing Postman inputs, three Newman result bundles, CI workflow, derived summaries, and ten attested PNGs; canonical header coverage= 485/485 .	VALID	The screenshots were visually reviewed, renamed only after byte-identical SHA-256 comparison, registered with matching metadata, and curated without changing their pixels. Derived output preserves 59 = 16 PASS + 28 FAIL + 15 SPEC-GAP and does not turn evidence handling into a new runtime claim.	The student stated that the screenshots were personally captured and explicitly attested them. AI did not claim authorship of the screenshots or substitute its visual review for the student's attestation.

3R. AI-Assisted FR-15 Completion and Freeze Bookkeeping Artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user handoff prompt, verbatim excerpts: "FR-15 implementation, execution, evidence, Postman screenshots, curation, and validation are now complete." and "Treat FR-15 as complete for the API-testing workflow." The prompt required a canonical-data check, AI Audit bookkeeping, a final-submission handoff, validation, commit/push, and no final artifact generation. Tool: OpenAI Codex; exact model/chat time were not separately persisted. Date: 23/08/2026.	Verbatim artifact/result identifiers: submission-checklist.md checkpoint 50 records FR-15 = COMPLETE / FROZEN , 59 executed cases, 16 PASS, 28 FAIL, 15 SPEC-GAP, 0 HARNESS, 0 BLOCKED, ten attested screenshots, 485/485 header coverage, published Issues #17/#18, and the strict validator's 14 deferred errors; ai-audit-report.md is updated through this checkpoint.	VALID	Every terminal value was re-read from canonical registries, immutable run data, and actual fr15/ files. Normal derive/validation and diff checks pass, while strict validation remains honestly failing for deferred submission artifacts; no placeholder, rerun, or weakened rule was introduced.	The student explicitly froze FR-15, bounded this chat to handoff/audit bookkeeping, prohibited final-submission artifact generation, and preserved the self-drawn diagram and YouTube URL as human boundaries.

3S. AI-Assisted Final Deliverables Audit and Report Bundle

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user prompt, verbatim excerpts: "FINAL HW06 SUBMISSION COMPLETION"; "Re-read the authoritative HW06 requirements"; "For each requirement classify it"; and "Do not fabricate any missing human action or evidence." Tool: OpenAI Codex (GPT-5). Date: 23/08/2026; the exact chat timestamp is not exposed to the repository.	Verbatim artifact identifiers: main-report.md ; README.md ; submission-checklist.md ; github-repo-link.txt ; sources/hw06-requirements.md ; provenance/test-summary.json . The checklist records three complete API pipelines, a conservative self-assessment of 090/100, and the genuine human-only	VALID	The bundle was derived after re-reading official sections 5–15 and reconciling repository evidence. Counts match canonical data: 163 AI cases, 17 human-added cases, 159 executed cases, 56 failures, and six published bugs. It does not reopen frozen API workflows or claim missing human evidence.	The student required an evidence-first final audit, prohibited optimistic credit and fabrication, required exact artifact paths, and explicitly clarified that oral-defense selection is post-submission rather than a deliverable.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
	boundaries for group non-duplication and the student-drawn diagram.			

3T. AI-Generated Critique Artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user prompt, verbatim excerpts: "Create/finalize the required AI critique"; "Base the critique on the actual HW06 process"; and "Do not claim that I manually wrote text that AI produced." Tool: OpenAI Codex (GPT-5). Date: 23/08/2026; the exact chat timestamp is not exposed to the repository.	Verbatim artifact excerpt from <code>ai-critique.md</code> : "The lesson is that AI should expand the search space and structure traceability, while humans must own requirements, oracles, evidence, and final judgment." The artifact is 260 whitespace-delimited words including its heading and addresses strong oracles, duplicate/padding removal, state/security value, SEC-04/SEC-05 limits, coincidental PASS results, and human evidence control.	VALID	The critique is within the mandatory 200–300 word range and uses concrete FR-04, FR-08, and FR-15 evidence instead of generic commentary. It explicitly attributes the text to AI-assisted finalization and does not claim student authorship.	The student supplied the required critique topics, required project-specific examples, and prohibited false manual-authorship claims.

3U. AI-Assisted Git Evidence Artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user prompt, verbatim excerpts: "Read the actual Git history"; "Use real commit hashes and real commit messages"; and "Select meaningful commits ... instead of dumping every commit indiscriminately." Tool: OpenAI Codex (GPT-5). Date: 23/08/2026; the exact chat timestamp is not exposed to the repository.	Verbatim artifact identifiers from <code>git-commit-log.txt</code> : selection/specification commits <code>83c08aad...</code> and <code>5b667939...</code> ; generation commits <code>658aeeca...</code> , <code>42eb5f0f...</code> , and <code>785f5de7...</code> ; execution commits <code>9066a735...</code> , <code>d52736a4...</code> , and <code>9c62e113...</code> ; CI and curation commits through <code>3c9d8d3f...</code> .	VALID	Every listed full SHA resolves to a real commit and every subject/date is copied from Git history. The artifact maps meaningful workflow stages and excludes unrelated HW04/HW05 history.	The student required stage-oriented selection, prohibited invented or rewritten history, and required real hashes/messages.

3V. AI-Generated Generator Pseudocode and Skill Package

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user prompt, verbatim excerpts: "finalize the API-test-generator design/pseudocode presentation" and "Do not fabricate any missing human action or evidence." Tool: OpenAI Codex (GPT-5). Date: 23/08/2026; the exact chat timestamp is not exposed to the repository.	Verbatim artifact identifiers: <code>generator/pseudocode.md</code> ; <code>generator/agent-skill/SKILL.md</code> ; <code>generator/README.md</code> . The pseudocode separates source extraction, domain partitions, state, security, schema, deduplication, oracle limits, provenance, and the later human-controlled pipeline.	INCOMPLETE	The reusable Skill and pseudocode satisfy the AI-generated portions of section 7, but the mandatory student-designed/self-drawn diagram is absent. Producing that diagram with AI would violate the anti-cheat constraint, so the generator deliverable cannot be marked complete.	The student required truthful handling of human-only evidence. No student-drawn diagram file was supplied, so AI preserved the gap rather than generating or misattributing one.

3W. AI-Assisted Submission Build and Self-Containment Tooling

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user prompt, verbatim excerpts: "Make out/ fully independent and self-contained"; "Add or use a lightweight deterministic out/ self-containment check"; and "Do not weaken validators merely to make them pass." Tool: OpenAI Codex (GPT-5). Date: 23/08/2026; the exact chat timestamp is not exposed to the repository.	Verbatim artifact/result identifiers: test-cases.xlsx ; main-report.pdf ; ai-audit-report.pdf ; final validation record= validation-report.txt . Repository-local helper scripts generated and checked these outputs but are not dependencies of this standalone folder. The first pre-audit self-containment run reported PASS (0 errors, 0 warnings) ; workspace validation passed, and strict submission validation reduced from 14 errors to the two honest missing-diagram errors.	INCOMPLETE	The self-containment checker itself is sound: it checks local Markdown links, path escape patterns, report/image existence, forbidden databases, token/private-key signatures, counts, Issue records, Git hashes, CI links, and PDF/XLSX signatures, and it does not suppress the missing self-drawn-diagram requirement or modify frozen test outcomes. The accompanying curation normalizer was defective. Its workflow-path rewrite embedded the search string inside its own replacement, so it was not idempotent: each repeated curation run prefixed an already-rewritten URL again, and three runs left eight tripled <code>.../blob/hw06-api-testing/https://... links in ai-audit-report.md . validate-out.mjs</code> did not detect this because a malformed absolute URL is skipped by its local-link check. This verdict supersedes the VALID verdict originally recorded for this row.	The defect was found and repaired during the later takeover pass recorded in section 3X; no frozen test outcome, assertion, or count was affected, because only submission-facing link text was corrupted.

3X. AI-Assisted Submission Takeover, Defect Recovery, and Final Packaging Artifact

This row audits the takeover pass that resumed final-submission work after the previous tool reached its usage limit mid-pass. It audits one AI-produced artifact: the recovery and final-packaging change set. It does not restart, rerun, or re-verdict any FR-04, FR-08, or FR-15 test artifact.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Actual user prompt, verbatim excerpts: "You are taking over this HW06 repository from Codex because Codex hit its usage limit in the middle of the final-submission pass"; "Do not discard, reset, overwrite, or revert its uncommitted work"; "If an existing partial change is sound, continue from it instead of rebuilding it from scratch"; and "Do not weaken validators merely to make them pass. Fix real issues." Tool: Claude Opus 5 (<code>claude-opus-5</code>) via Claude Code. Date: 23/08/2026; the exact chat timestamp is not exposed to the repository.	Verbatim artifact/result identifiers: repaired eight corrupted workflow links in <code>ai-audit-report.md</code> ; made the workflow-path rewrite in the curation script idempotent via a negative-lookbehind guard plus a collapse pass, verified by two consecutive curation runs producing the identical MD5 <code>0d44b80575622c6b8a55434dfeae43f2f</code> ; added <code>ai-critique.pdf</code> and registered it in <code>README.md</code> , <code>submission-checklist.md</code> , and the self-containment validator; corrected an empty code span in <code>submission-checklist.md</code> . Deterministic results after the pass: workspace validation PASS (0 errors, 0 warnings) , self-containment PASS (0 errors, 0 warnings) , strict	VALID	The pass preserved the previous tool's uncommitted work instead of rebuilding it, and changed only what repository evidence proved to be broken. The recovered defect was a genuine correctness fault, not a style preference, and the fix is verified by an idempotence check rather than by inspection. The two remaining strict-validation errors were deliberately left failing because they encode the genuine student-authorship requirement of section 7; no validator was relaxed and no diagram was fabricated. Adding <code>ai-critique.pdf</code> resolves the section 14 phrase "AI Critique and AI Audit Report (Markdown + PDF)" in the	The student directed the takeover, prohibited discarding the in-progress work, prohibited weakening validators, and required that any genuine human-authorship requirement be reported as pending rather than satisfied by AI. The self-drawn generator diagram and the demonstration video URL remain the student's to supply.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
	submission validation FAIL (2 errors) for the still-missing self-drawn generator diagram.		reading that requires a PDF for both documents.	

4. Summary of AI Accuracy

Metric	Count	Percentage
Total AI-generated artifacts audited	191	100.00%
VALID	132	69.11%
INVALID	21	10.99%
INCOMPLETE	38	19.90%

Percentages are independently rounded to two decimal places and total 100.00%. The section 3W tooling row was reclassified from VALID to INCOMPLETE during the section 3X takeover pass after a real idempotence defect was found in the curation normalizer; it is re-verdicted in place rather than double-counted as a second artifact.

The FR-04 subset remains unchanged: 47 AI test artifacts comprising 21 VALID, 16 INCOMPLETE and 10 INVALID, plus six later major artifacts comprising five VALID and one INCOMPLETE CI smoke artifact. The 37 usable AI-origin FR-04 cases are the 21 VALID cases plus the 16 INCOMPLETE cases after confirmed corrections.

The FR-08 subset adds 61 artifacts: 57 test candidates comprising 40 VALID, 11 INCOMPLETE and 6 INVALID, plus one VALID Postman implementation, two VALID published GitHub Issues, and one INCOMPLETE CI demonstration. The 51 usable AI-origin FR-08 cases are the 40 VALID cases plus the 11 INCOMPLETE cases after the student's approved corrections; with the five student-selected extensions the executable FR-08 suite is 56 cases.

The FR-15 subset adds 71 artifacts: 59 test candidates comprising 49 VALID, 5 INCOMPLETE and 5 INVALID, plus ten VALID and two INCOMPLETE AI-assisted workflow artifacts. The 54 usable AI-origin FR-15 cases plus five student-selected HUMAN extensions form the 59-case executable suite.

5. Conclusion – When should AI be used (or not)?

AI is useful for decomposing an authoritative API specification into candidate partitions, state checks, security ideas, schema assertions, structured bug reports, and reproducible automation. It must remain an auditable assistant rather than an undocumented source of requirements or human approval. AI should not invent HTTP statuses, validation rules, response schemas, confirmed bugs, execution evidence, or green CI compliance when authoritative evidence says otherwise. Human review is necessary to separate official contract cases from exploratory robustness cases, isolate faults, define deterministic inputs, reject duplicates, constrain security claims, and expose assignment conflicts. In this audit, both CI demonstration workflows are deliberately classified INCOMPLETE even though their real sample runs work correctly, because a one-case green run does not prove that all canonical cases pass. The FR-08 phase adds two further lessons. First, capturing the exact prompt, tool and timestamp at generation time turned the audit from a reconstruction problem into a reading problem: every FR-08 row quotes a prompt that existed before the output. Second, a passing test is not automatically evidence: because the implementation never derives the checkout total from the cart, the FR-08 cases whose client value happened to equal the cart total pass without proving anything, and that limitation is recorded next to the results instead of being presented as coverage.

6. Mandatory Disclosure

I use AI tools for the following tasks: API-selection analysis, scoped API-specification documentation, generation and correction of the original 47 FR-04, 57 FR-08, and 59 FR-15 test candidates, Postman/Newman automation support, drafting the six published GitHub Issues, evidence curation, completion bookkeeping, implementing and reviewing GitHub Actions workflows, and generating the final report, checklist, critique, generator pseudocode, Git evidence selection, Excel/PDF export tooling, and self-containment validation. The AI tools were Codex for FR-04 and the FR-15/final-submission workflow, Claude Opus 5 (`claude-opus-5`) via Claude Code for FR-08

and FR-15 generation batches B1-B5, OpenAI Codex (GPT-5) for FR-15 batch B6 and final curation, and Claude Opus 5 (`claude-opus-5`) via Claude Code again for the final takeover, defect-recovery and packaging pass recorded in section 3X. I reviewed and modified the API-workflow artifacts against the official HW06 requirement, the official EShop API specification, the EShop README, canonical registries, and real execution evidence. The final submission text and tooling are AI-generated and are not claimed as manually written by me. Seven FR-04 extension cases, five FR-08 extension cases, and five FR-15 extension cases are student-selected or student-authored and are not counted as AI-generated test artifacts; for FR-08 and FR-15 the underlying ideas were AI-proposed and the selection, modification and approval were mine, which is stated in each `humanExtensionRationale`. Real Newman output, GitHub pages, CI runs, and screenshots were captured from actual systems and were not fabricated by AI. All ten FR-15 screenshots, including the three Postman Desktop screenshots captured by me, carry my explicit attestation. The required student-designed/self-drawn generator diagram has not yet been provided and must not be generated or falsely attributed to AI.

Signature

Field	Value
Student name (printed):	Nguyen Bao Duy
Student ID:	23127179
Class / Cohort:	23KTPM2
Course:	CS423 / CSC13003 – Software Testing
Instructor:	Course lecturers and TAs listed in the official HW06 requirement
Date:	22/08/2026
Signature:	Not required by the official HW06 submission specification (electronic submission)

References

- [AI-02] - FIT@HCMUS - AI Audit Report_En.docx (2).md, FIT@HCMUS AI Policy Templates – 2026 v1.0.
- Official HW06 requirement: `sources/hw06-requirements.md`.
- Official SUT API specification: `sources/eshop-api-specification.md`.
- EShop requirements and security requirements: `sources/eshop-requirements.md`.
- ISTQB Foundation Level Syllabus, latest version.
- Hardman, P. (2025). A Post-AI Learning Taxonomy.

Completion note

This report records 191 audited artifacts through the final-submission takeover interaction: the previous 185 API-workflow artifacts, five final report/critique/Git/generator/tooling artifacts, and one takeover/defect-recovery artifact.

FR-04 contributes 53: 47 original AI-generated test-case artifacts, two AI-assisted correction bundles, two published GitHub Issue artifacts, and two CI workflow artifacts. Its seven HUMAN extension cases remain separately recorded in `provenance/test-cases.json`. Its two bugs are published as GitHub Issues `#13` and `#14`, its canonical full suite executes 44 cases with 35 PASS and 9 FAIL, and strict "all API test cases passing" compliance remains PARTIAL because the unchanged suite retains confirmed SUT-defect failures and one specification-gap observation.

FR-08 contributes 61: 57 original AI-generated test-case artifacts, one Postman implementation artifact, two published GitHub Issue artifacts, and one CI demonstration artifact. Its five extension cases were selected and approved by the student from an AI-proposed shortlist and are not counted as AI test artifacts. The canonical run `RUN-20260823080014785-fr08-canonical-full-suite` executed all 56 executable cases with 37 PASS and 19 FAIL and preserved its original exit code 1; every failure maps to the two bugs published as GitHub Issues `#15` and `#16`. All ten FR-08 evidence items are human-attested and curated under `fr08/`. Strict all-tests-green CI compliance is not claimed for FR-08: its CI sample runs one case and no FR-08 canonical full-suite workflow was built.

FR-08 provenance is complete: every candidate stores its verbatim bounded prompt, batch id, actual tool and real generation timestamp, so no FR-08 prompt is reconstructed from an output. FR-04 generation prompts were not captured at the time and remain explicitly marked as such. No unsupported prompt, time, citation, runtime result, or human decision is invented anywhere in this report.

FR-15 contributes 71 artifacts: 59 original AI-generated test-case artifacts and twelve later AI-assisted workflow artifacts. Its five student-selected HUMAN extensions are not counted as AI test artifacts. The canonical run `RUN-20260823102155874-fr15-canonical-full-suite` executed all 59 cases with 16 PASS, 28 FAIL, 15 observational SPEC-GAP outcomes, 0 HARNESS and 0 BLOCKED. Every authoritative failure maps to one of two published root defects: GitHub Issues `#17` and `#18`. All ten FR-15 evidence screenshots are student-attested and curated under `fr15/`; all 485 canonical requests carry `X-Student-Id: 23127179`.

This report is current through this final-submission curation interaction. The student-designed/self-drawn generator diagram, external group non-duplication confirmation, and final YouTube URL remain human-provided boundaries. If the student later supplies the diagram or video and AI assists with integration, that later interaction must be appended before the ZIP is frozen.